The second most common reason for creating a home or office network is to provide access to shared files. In many offices, a central computer or server is used to store shared files and anyone who needs access to these files, can access them through the server.

Homes and offices also use networks to share hardware, such as printers and scanners. This allows many people to use a single printer or scanner without too much hassle or causing any problems.

WHAT IS THE CLOUD?

https://www.youtube.com/watch?time_continue=30&v=gu4FYSFeWqg

NETWORKING TO SUCCESS!

At some point you must have come across the terms *cloud*, *cloud computing*, or *cloud storage*, but what exactly is this cloud?

In simple words, the cloud is the internet and refers to all the things you can access remotely through the internet. If a file or folder, or pretty much anything that is stored on the cloud, is stored on internet servers and not on your computer's hard drive.

People all over the world use the cloud, because it is reliable and convenient. It is easy to store and share files, as well as back-up data. If, for example, you have used an email application, such as Gmail or MWeb mail, you have already used the cloud!

The nice thing about the cloud is that even if your computer crashes, you can still access your information if you have previously backed it up on the cloud.

Have a look at the video link on the left to understand more about cloud computing, it is worth a watch!

When we think of resources, remember these are not just restricted to your computer, but can also extend to real-world resources, for example, people and money. Should you encounter a problem, you can get advice from experts all over the world by asking for advice on the internet. Banking sites, investment sites and **crowdfunding** websites, such as GoFundMe.com, can provide you with access to money.

ORGANISING INFORMATION

Computer networks not only provide access to information; they also help people to organise information. This could be something small, such as a Google Calendar event where people work together to organise a single event. It can be something complex, such as Google Maps that gives people driving directions in more than 240 countries and covers over 64 million kilometres of road (that is about 83 trips to the moon and back).

By recording information in a central location on a network, people can work together to organise it in a way that is useful and understandable.

CONNECTING PEOPLE

Another important use of networks or telecommunication networks is to connect people. This can be through emails, video calls on your computer, social networks such as WhatsApp or other instant messaging services (IMSs). Using computer networks makes this possible and allows people from anywhere in the world to communicate with one another.

Figure 9.3: *Using the network to communicate*

ACCESS TO ENTERTAINMENT

A final important use of networks is to give people access to entertainment. In the past, entertainment was an expensive luxury that only a few people could have. Today, things are different; the average person with an internet connection can watch and listen to a wide variety of films and albums on the internet. In fact, internet access provides access to an almost unlimited amount of entertainment; from books to read, music to listen to, watching the news, or chatting with friends.

ADVANTAGES AND DISADVANTAGES OF NETWORKS

Networks also pose potential advantages and disadvantages. The following table looks at the risks and benefits that come with networks.

Table 9.1: *Comparison of the advantages and disadvantages of connecting to a network*

ADVANTAGES	DISADVANTAGES
• **Makes communication easier:** People can communicate from all around the world through the network. • **Information transfer:** Data and information can be easily transferred or copied from the one computer to another. • **Access to hardware devices:** In a network, access to a single printer or scanner is easily available to many people at the same time. • **Software installation:** Software applications can easily be installed or upgraded from one central location. • **Central storage:** Information and data can be stored in one place (central location).	• **Online crimes:** Can expose a person to online crimes, such as identity theft, credit-card theft and scams. • **Viruses:** Viruses can easily be transferred from one computer to another through the network. • **Security and privacy concerns:** Since computers are all connected in a network, people can try to access private and restricted information using the network. • **Maintenance:** A network administrator has control over the network, and administers and manages usage.

Everything has its advantages and disadvantages; it is just the way life goes. However, the table on the previous page is just a list of potential advantages and disadvantages; the way in which you use the network is the only way you get affected, positively and negatively. For example, if you want to learn how to bake, speak a new language or even build your own treehouse, there are always free resources to teach you a new skill. But at the same time, the downside is that the internet also has a dark side. So, what you get from a network depends on what you are looking for and your ability to use it.

WHERE DOES THE INTERNET COME FROM?

https://www.youtube.com/watch?time_continue=17&v=jKA5hz3dV-g

THE INTERNET: THE NETWORK OF NETWORKS

We have spoken a lot about networks and how networks enable telecommunication, but what exactly is the driving force behind that? The internet is a global network made up of many, many computers (we are talking about billions of computers) and other electronic devices. You can access almost any information, communicate with anyone anywhere across the world and do much more, just by connecting a computer or electronic device to the internet.

There are also a lot of other things that you can do on the internet. As you know, one of the best things about the internet is how quick it is, and how quickly you can communicate with anyone anywhere in the world. You can email and use social media, you can pay bills, do online shopping, listen to music, meet new people, or even learn a new skill.

UNIT

9.3 Social implications: Networks

There are legal, ethical and security aspects you should take note of with regards to networks. These include viruses, licensing contracts, adhering to the user policies and ownership of electronic material. In this section, we will look at each of these aspects in some more detail.

VIRUSES

This is anyone's worst nightmare. Without a proper antivirus program installed, you will be prone to getting a virus. Viruses can be transferred through flash disks, or through other computers in a network. Antivirus programs help to protect your software, data and PC from possible threats.

Read through the following case study and discuss the questions in small groups.

Case Study 9.1 Viruses are dangerous

Viruses can be very dangerous. In 2000, there was a virus called the "ILOVEYOU virus". Although, by today's standards, it is a pretty silly virus, it still is one of the most well-known and destructive viruses of all time. It even made the Guinness World Record for being the most "**virulent**" virus of all time. It overwrote both system and personal files, and spread itself over and over again.

The virus was a worm that was downloaded by clicking on an email attachment called "LOVE-LETTER-FOR-YOU.TXT.vbs". Although people did not know the person sending the email, or the fact that it made headlines around the world, people still clicked on it, activating the virus and attaching it to all their emails.

In the end, the two programmers who developed the virus were caught. Unfortunately, they were set free as there were no laws against writing **malware** at that time.

Discuss the following in pairs.

1. Will you open an email sent to you from someone you do not know? Motivate your answer.
2. How will you know that something (an email, website, etc.) is dodgy? What would you do in such a case?
3. What can you do to ensure that you do not get a virus like that?

LICENSING CONTRACTS

An organisation cannot just share software over a network, it must purchase a network licence; otherwise, it is illegal. A network licence lets more than one user at a time access the software on the server.

Depending on the number of computers attached to the network, the network licence fee will be different. A legal agreement that lets users install software on a number of computers is called a **site licence**.

USER POLICIES

Organisations, such as schools and businesses, normally have an *acceptable computer usage policy* in place. This policy stipulates how computer equipment should be used and prevents anyone from accessing restricted information or data on the network. If a person breaks the rules, it is regarded as a serious offence and the person could get into a lot of trouble.

OWNERSHIP OF ELECTRONIC MATERIAL

Accessing data that is not your own is regarded as a criminal offence. For example, you should not change or access the data on the computer of another network user without getting his or her permission. **Plagiarism** is when you illegally copy information from the internet, or from any other published material and say that it is your own, unless you obtained permission to use it.

Something to know

Believe it or not, plagiarism is actually very old; it just was not frowned upon then as it is now. Shakespeare is known to have "borrowed" a lot from other writers. However, back in the day, plagiarising was actually seen as a compliment.

Case Study 9.2 Plagiarism in music

The "Blurred Lines case" made the news when Pharrell Williams and Robin Thicke were accused of copying Marvin Gaye's music to create "Blurred Lines". This was the most popular song in 2013. Marvin Gaye's children were awarded US $7.4 million since the jury was in favour of their claims.

Answer the following questions on your own.

1. Why was it seen as plagiarism? What did they do wrong?
2. What do you think about plagiarism – should the law continue working against plagiarism? Why or why not?

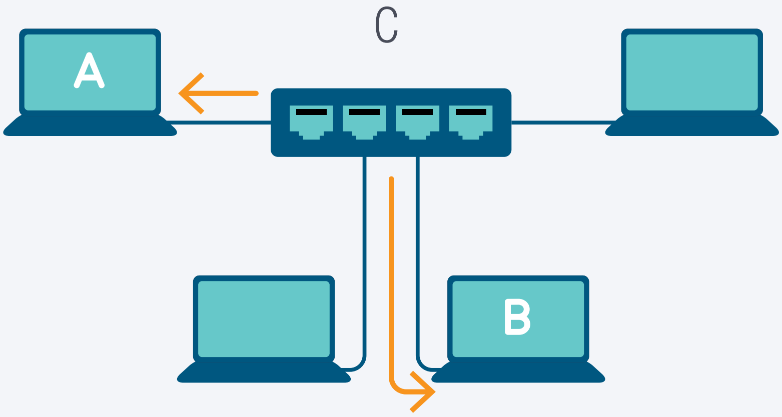
Activity 9.2

Miss Furry wants to network the computers and other hardware devices in her office. She shares a printer with three other people. She was told she would need a switch, router and a **modem**.

1. Define a computer network.
2. Answer the following questions with reference to the scenario above:
   a. Give two advantages of creating a network for Miss Furry.
   b. List one type of hardware device that she can connect to the network.
   c. Is this hardware (your answer for question b above) an input or output device? Motivate your answer.
   d. Briefly explain how Miss Furry can use the network to communicate to others.
3. Explain what the internet is.
4. An organisation cannot just share software over a network.
   a. Provided that this statement is true. What can an organisation do to share software?
   b. Briefly give a description of a network licence.
   c. Explain what a site licence is.

REVISION ACTIVITY

1. What is a computer network? (4)
2. Your school is planning to network its 25 stand-alone computers.
   a. List two advantages to the school of installing a network. (2)
   b. List two disadvantages to the school of installing a network. (2)
   c. Apart from a network server and cabling, what other hardware would the school need when installing the network? (2)
3. The diagram below shows a simple network. Use the diagram to answer the following questions:



   a. What are A and B? (2)
   b. What is device C? There are two possible answers. List both of them. (2)
   c. Which device is receiving information? (1)
4. "The internet is an example of a computer network." It this statement true or false? (1)
5. Reinet is doing research for her PAT. She finds an interesting website on the internet. When she prepares her final presentation, she copies large sections of text from the website and pastes them into her presentation.
   a. Is Reinet allowed to copy sections of text into her presentation and pretend that she wrote it herself? Give a reason for your answer. (2)
   b. Give one word for what Reinet has done. (1)

TOTAL: [19]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	DID YOU ...	YES	NO
1.	Understand the different concepts that are used in networks.		
2.	Learn about the different uses of networks.		
3.	Learn about the advantages and disadvantages of networks.		
4.	Understand the legal, ethical and security aspects regarding networks.		

TERM 3

CHAPTER 10 TYPES OF NETWORKS: PAN/HAN

CHAPTER OVERVIEW	
Unit 10.1	PAN/HAN
Unit 10.2	Creating a PAN/HAN

At the end of this chapter, you should be able to:

- Discuss the different types of networks.
- Identify the advantages and disadvantages of PAN.
- Distinguish between a modem, router and switch.
- Identify the hardware and software required to connect to the internet using a PC.

INTRODUCTION

In the previous chapter, you learned about the different concepts used in networks. In this chapter, we will look at PANs and HANs, and how they are created, their advantages and disadvantages, as well as the different network devices required to create a network.

There are different types of networks. For ease of reference, we have arranged them in the following list, according to their size – i.e. from the smallest to the largest network.

1. Personal area network (PAN) and home area network (HAN) as the names state, are both used in small areas, such as home or small office environments.
2. **Local area network (LAN)** or **wireless local area network (WLAN)** are used in bigger offices and in schools.
3. **Wide area network (WAN)** connects more than one LAN in different places, for example, cities or buildings, into one big network.

Figure 10.1 shows the different types of networks.

Networks can be categorised according to their size and distance covered. As you can see, the WAN is the largest network with the internet being the most popular WAN.

Figure 10.1: *Types of computer networks*

Although all these networks may sound the same, they are quite different. In Grade 10, you will focus on the difference between a HAN and a PAN.

WIRED AND WIRELESS NETWORKS

Before looking at HANs and PANs, you first need to understand the difference between a wired and a wireless network.

A *wired network* is a common type of network and uses **ethernet cables**, or fibre-optic cables to transfer data between computers that are connected to the network.

Wireless networks allow many devices to connect to the same internet connection, as well as to share files and other resources. It refers to a network where the devices are connected via radio- or microwaves, and not through physical cables.

Something to know

The internet is the biggest network in the world. The funny thing is that no one actually owns it and at the same time, lots of people own it. Confusing, right? If you think of the internet as one entity, no one owns it, but at the same time, the internet is made up of small little parts and each of these parts has an owner. From this perspective, lots of people and organisations own the internet.

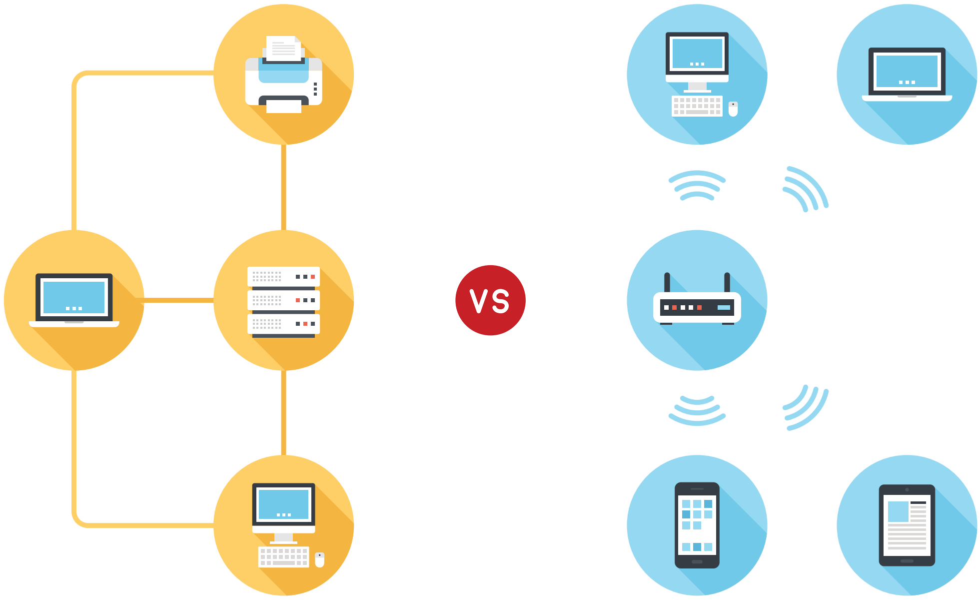


Figure 10.2: *Comparison between a wired network (left) and wireless network (right)*

HANS AND PANS

HAN

As the name suggests, a HAN is a very small network that usually covers a single home. Any device that is connected to this network will be able to share resources, for example the internet, smart appliances, printers, smart meters and even some security systems.

Figure 10.3: *Example of a home network, where a computer, Smart TV, and a phone are all connected to one small network*

It allows the computers on the HAN to communicate with each other directly and to transfer files between the computers, send messages, or even play LAN games. HANs can be both wired or wireless networks.

GETTING THE MOST OUT OF YOUR SMART DEVICES
Jenny stays upstairs in a two-storey house. The printer is in the study downstairs, and even though it is good exercise to go downstairs with a flash disk and connect to the main computer to print something, it is still time consuming. So, Jenny's dad made a plan. Since they have a wireless printer, it has the capability of connecting directly to the home network. He installed and connected the printer wirelessly to each computer in the house. Now Jenny can just print whatever she needs from her room and fetch the printed document later on. This is an example of how convenient a home network is.

PAN

A PAN is like a HAN, but even smaller! It refers to a network built for a single person and contains all the devices connected to this network. This can include all devices connected on most networks, for example a computer, smartphone and printer. It can also include personal devices, such as Bluetooth keyboards, Bluetooth earphones and smartwatches. Unlike LANs and HANs, all devices on a PAN do not have to be connected using an **IP address**. Some devices can be connected over a Wi-Fi network, while others might connect using Bluetooth, or even USB.

For example, when the iBooks application on the computer knows which page you last read, it is because of the PAN. Those devices are connected and "talk" to each other; they are basically synced to each other. Another example is the fact that your smartwatch can tell you of any messages, emails or incoming calls while your phone is charging.

The goal of a personal network is to make the user's life easier by allowing individual devices to communicate directly with each other.

THE FUTURE OF WEARABLE DEVICES
Now that you understand a little about PANs, let's see how this goes with fitness. Whether you are wearing an Apple watch, a FitBit, or heart-rate monitor, these devices can all connect and speak to each other. They all measure different things; from the calories you burn, to your sleep patterns and how many steps you took in a day, and much more. These measurements are then communicated to your phone or smartwatch, creating a little report on your fitness statistics. This type of PAN for the fitness industry was created by Dynastream and is called ANT.

PAN VERSUS HAN

As can be seen in the following table, both networks have various advantages and disadvantages.

THE ADVANTAGES AND DISADVANTAGES OF PAN

The following table shows the different advantages and disadvantages of a PAN.

Table 10.1: *Advantages and disadvantages of PAN*

ADVANTAGES	DISADVANTAGES
Space saving: PANs do not require extra wire or space. To connect two devices, for example a wireless mouse and a laptop, all you need is Bluetooth.	**Distance:** Because there is a limitation of 10 metres, this makes it difficult for sharing information and data over long distances.

... continued

ADVANTAGES	DISADVANTAGES
Easy to use: It is easy to use and no complicated setup is required. **Portable:** You can move these devices easily as they do not have a whole bunch of wires that come with them. **Reliable:** This type of network is reliable and stable for the devices within ten metres of the network. **Secure:** Files and information shared on this network can only be accessed by authorised people. **Synchronisation:** You can synchronise as many computing devices, where you can download, upload, or exchange information between the devices.	**Costly:** PAN only makes use of digital devices, and these types of devices are expensive, for example, laptops, smartphones, smartwatches and many more. **Slow data transfer:** The use of Bluetooth to transfer files has a slow transfer rate compared to other types of networks, such as LANs. **Health issues:** In some instances, PAN uses microwave signals in digital devices that can have a negative effect on the human brain and body.

ADVANTAGES AND DISADVANTAGES OF HAN

The following table shows the different advantages and disadvantages of a HAN.

Table 10.2: *Advantages and disadvantages of HAN*

ADVANTAGES	DISADVANTAGES
Accessibility: Allows several users to be connected to the same internet connection. **Resources:** Resources, such as printers, faxes and files can be shared over the same network. **More economical:** Because several users can use the same hardware and internet, this reduces the costs of these things.	**Internet:** Sometimes, if someone is downloading a big file from the internet, this can slow down the internet speed drastically for other users. **Security:** Sometimes, if your home network is not secure, a person that lives close by can have access to files and folders on your network, so a password is necessary. **Costly:** Buying all the equipment required, depending on the number of computing devices that must be connected to the home network, could be costly.

UNIT

10.2 Creating a PAN/HAN

In the past, home networks were not so popular. Most families did not need, or could not afford more than one computer. Today, this is not the case; people use their computers for school work, shopping, downloading videos and music, watching movies, instant messaging and so on.

So, having one computer in a household is not enough anymore and multiple computers or computing devices are becoming more of a necessity than a luxury.

REQUIREMENTS

There are several options to look at when creating a network in your household. This section will look at what to keep in mind when creating home networks and what types of hardware are needed to create and protect your home network.

INTERNET INFRASTRUCTURE

Any computer that is connected to the internet is part of a network. This can be from the 100 computers connected in an office, to just the one in your home. For example, at home you can connect to the internet using a modem. The modem dials a local number that connects to an **internet service provider (ISP)**. The ISP is the term used for a company that provides you with access to the internet; this could be from your computer or even your smartphone.

The ISP makes the internet a reality. Suppose you have a brand-new computer with a built-in modem and a router to connect to the network. However, without an ISP subscription, you will not be able to connect to the internet.

WHAT YOU NEED FOR A PAN
To create a PAN, you need a minimum of two computing devices, for example a PC and a smartphone. You will also need a communication channel, which can be wired or wireless to transfer information between the different devices. FireWire and USB are examples of a wired PAN; while WPANs generally use Bluetooth, or even infrared technology. PANs can only transfer information between devices that are close to each other instead of sending it over the internet (WAN). These networks can be used to transfer files, such as music, photos, videos and calendar appointments. The easiest way is to transfer the data through a PAN. You can use a USB cable to connect the phone to the laptop. Then follow the **prompts** shown on the laptop, access the files from the phone's storage and copy them onto the laptop. The user can now easily access information. You can also use another way to transfer data. You can sync both the laptop and phone using Bluetooth, and then sending data to the laptop, or vice versa.

How data is transferred

Watch this video to see how data can be transferred from the phone to the laptop: https://www.youtube.com/watch?v=gjsiD9i8l_8

WHAT YOU NEED FOR A HAN

HANs can be both wired or wireless networks. In a typical HAN setup, a router is used as the central point of the network. This router allocates IP addresses and provides internet connectivity to all devices on the network. Any device in the household can then connect to the router, by either connecting to its Wi-Fi network, or by connecting to it with an ethernet cable. Once connected, the devices automatically have access to the internet and network resources.

NETWORK DEVICES

In most cases, a wired or wireless home network requires only the computing devices, modem and router. Obviously, this depends on your requirements; the more complex you want the network to be, the more money you will be spending on the equipment required. In this section, we will discuss the different network equipment required to create a home network.

A **network adapter**, also known as a network interface controller (NIC), is a piece of hardware that can be added to a computer, allowing it to connect to a network. These days, most computers and laptops have a network adapter built into the motherboards, which makes setting up to the internet much easier.

There are three important types of networking equipment that can look very similar from the outside – i.e. the modem, router and switch. These devices can have ports for different types of cables, such as ethernet cables or digital subscriber line (DSL) cables. However, each has their own different function, as shown in the following table:

Table 10.3: *The different network equipment*

TYPE	FUNCTION	IMAGE
Modem	The function of a modem is to connect a computer or network to the internet.	
Switch	The function of a switch is to connect many computers on the same, internal network.	
Router	The function of a router is to organise and route data on and between networks. This may include routing data from a home network to the internet (such as a modem) and connecting many computers to the same network (such as a switch).	

Today, home routers cost roughly the same as home switches and modems. Therefore, for most setups, it is easier to simply purchase a router that can serve more than one function. However, for larger businesses, specialised equipment, such as dedicated modems and switches, may be better suited to the job.

CONNECTING TO THE INTERNET

To connect to the internet, there are software and hardware requirements that must first be fulfilled. This section will look at those requirements and soon you will be able to create your very own home network with internet!

SOFTWARE REQUIREMENTS

The software you need to connect to the internet is an operating system, a **web browser**. A web browser, such as Firefox or Google Chrome, is used to display pages that you visit on the internet. Web browsers often come with their own operating systems.

HARDWARE REQUIREMENTS

Connecting to the internet is quite simple. You will need the following hardware equipment to connect to the internet:

- A computing device, such as a computer or a smartphone
- A communication channel, such as a telephone line and ethernet cable
- A modem or router to connect to the internet

Example 10.1 How to set up a Wi-Fi network

Setting up a wireless router is quite simple. If you have purchased your router from an ISP, you will probably have an internet plan with them, and your router will come with all the things you need to connect to the internet, as well as a **subscriber identity module (SIM)** card.

Figure 10.4: *Home Wi-Fi network*

… continued

Example 10.1 How to set up a Wi-Fi network *… continued*

1. Insert the SIM card into the SIM slot of the Wi-Fi router.
2. Connect the router to the power supply. With some routers you might have to screw the two external antennae to the router.
3. Take note of the information that is on the label at the back of the router.
4. Plug the LAN cable into a port of the router and the other end of the cable into the LAN port of your computer.
5. Switch on the router by pressing the *Power* button.
6. Switch on your computer.
7. The power light will be on and will turn a specific colour. This means you are ready to install the software.
8. Follow the prompts on your computer.
9. The Wi-Fi light will be a specific colour to show that the Wi-Fi is enabled.
10. The strength bar on the router will show how strong the signal is.

After you have set up your Wi-Fi network and configured your router, you are ready to connect to the Wi-Fi. This procedure may differ, depending on the router and computing device.

To connect to the Wi-Fi network, you can do the following:

1. Click on your computer's network settings and search for nearby Wi-Fi networks.
2. Choose your network and enter the password you just created.
3. If the connection is successful, open your web browser and type in *www.google.com*. If the page loads, your Wi-Fi connection is working properly.

Something to know

Do not worry if you want to connect a computer that does not have built-in Wi-Fi connectivity. You can purchase a Wi-Fi adapter that plugs into your computer's USB port.

Creating a WAN

Create a real-life video of creating a WAN. It can be made step by step, with lots of jump cuts:

1. Get a Wi-Fi router.
2. Plug in the Wi-Fi router.
3. Get the Wi-Fi router name and password.
4. Connect the notebook using the name and password.
5. Open the Wi-Fi router *Settings* page.
6. Set a new Wi-Fi router name and password.
7. Connect to the new Wi-Fi router.

You can use the following link as a guideline: https://i.imgur.com/SqmxiZL.gifv.

Activity 10.1

The Jackson family is made up of four people. The two children, Tristan and Jess, stay upstairs, and the parents have a small work-from-home office. The children always have to go downstairs to print documents, or even just use the internet. A friend recommended they set up a network.

1. Differentiate between a PAN and HAN.
2. Briefly explain what a network adapter is.
3. What type of network is best for this situation so that everyone has access to the internet?
4. What three hardware devices are required for this connection?
5. What is a router? Explain its function.
6. Answer the following questions regarding the scenario:
   a. What type of network is best suited in this context? Motivate your answer.
   b. Explain two advantages and disadvantages of the answer you gave in (a).

REVISION ACTIVITY

1. Explain the difference between a HAN and a PAN. (4)
2. List one advantage and one disadvantage of a HAN. (2)
3. List one advantage and one disadvantage of a PAN. (2)
4. Three common network devices are modems, switches and routers. Match the description of the device to its name. Write down only the name of the device:
   a. Used to connect computers or networks to the internet (1)
   b. Use to organise and route data on and between networks (1)
   c. Used to connect many computers on the same network (1)
5. Write down the letter that matches the correct or best answer. Which of the following devices is most likely to have a built-in network adaptor?
   A. Microwave
   B. Refrigerator
   C. Telephone
   D. Sewing machine (1)
6. What communication medium is most likely to be used in a PAN? (1)
7. Sifiso has a desktop computer with the Windows 10 operating system installed on it. He has installed on it the Google Chrome browser on his computer. He wishes to connect his computer to the internet. He already has a 24-month contract with Telkom.
   a. Does Sifiso have the correct software to connect to the internet? (1)
   b. What additional hardware device, other than his computer, will Sifiso need to connect to the internet? (1)

TOTAL: [15]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	DID YOU ...	YES	NO
1.	Understand the difference between the different networks.		
2.	Learn what the wired and wireless networks are.		
3.	Understand the advantages and disadvantages of PAN.		
4.	Identify the computing devices required for a PAN.		
5.	Learn how to connect to the internet.		

CHAPTER 11

THE INTERNET AND WORLD WIDE WEB

CHAPTER OVERVIEW

At the end of this chapter, you should be able to:

- Explain what the internet and world wide web (WWW) are.
- Understand the concepts used in the Web.
- Differentiate between a website and web page.
- Understand what a web address is and the different elements that it consists of.
- Identify the different types of websites.
- Use the web browser.
- Open, close and switch between tabs.
- Use a search engine to browse the internet.
- Understand what uploading and downloading are.

INTRODUCTION

The internet has become increasingly important in everyday life for people all over the world. It is the biggest network made up of billions of computers and other computing devices, such as smartphones, tablets, laptops, etc. The internet allows us to communicate with anyone across the world and access almost any type of information we need.

The biggest trend now, thanks to the internet, is social media. So, before you post anything new on Instagram, let's learn more about the internet and social media.

Social media and other interactive, crowd-based **communication platforms** reached new heights at the beginning of the 21st century. This has resulted in people being more up to date both in terms of their own lives and in the lives of others.

In this chapter, we will look at what the internet and world wide web (WWW) are, the different types of websites, as well as browsers and how to do some basic browsing.

The internet is a world-wide system of computer networks that are connected to each other. These networks connect with each other using cables, telephone lines and communication satellites.

When a computer is connected to the internet, it allows a person to access emails, music videos, pictures and any other relevant information. If a person is connected to the internet, this person is said to be working **online**.

INTERNET ADDRESS

Have you ever wondered about the fact that if there are 250 billion emails sent every day from our smartphones, PCs or laptops, how exactly do they go to the correct place? The answer is that every device connected to the internet receives a unique IP address. Whenever a message is sent over the internet, it is sent to a device's IP address. An IP address consists of four sets of numbers, which are separated by dots, as shown below:

Figure 11.1: *An example of a South African IP address*

WHAT IS THE INTERNET?

Watch the following interesting video to listen to what the co-inventor of the internet says:

https://www.youtube.com/watch?v=Dxcc6ycZ73M

Guided Activity 11.1

1. To see what the IP address is of the computer you are currently working on, do the following:
   a. Open a web browser. Ask your teacher to show you how to do it if you do not know.
   b. Go to the Google search engine.
   c. Type in "What's my IP?" in the search box.
   Your IP address will be displayed at the top of the search results.

As South Africa only has **dynamic IP addresses**, the IP address will change every time you connect to the internet.

UNIT

11.2 The world wide web

The world wide web (WWW) is a part of the internet where documents and other resources can be accessed. The WWW is often called "The Web".

The internet and the WWW are often confused. The internet is, in fact, the biggest network in the world; while the WWW is a collection of documents and other resources that you can browse, or access, through the internet.

Most resources are **websites** that can consist of text, pictures, audio clips, video clips, animations, etc. After connecting to the internet, you can browse websites using a type of application called a **web browser**.

You will learn how to browse the internet, as well as how to navigate to the different websites using **uniform resource locators (URLs)** and tabbed browsing.

IMPORTANT CONCEPTS USED WITH THE WORLD WIDE WEB

There are some important concepts that you should learn in order to understand the Web better. Some of these concepts will be explained in more detail in the sections to follow.

A **web server** is a computer that hosts a website, for example if you want to access www.wikipedia.org, the web server receives this request and uses **hypertext transfer protocol (HTTP)** to format and then present the website to you.

A website is a collection of web pages, for example Wikipedia is a website that has billions of web pages about different topics and articles. A web page is a single page of **hypertext mark-up language (HTML)** text, which can display text, media, images, or interactive material, such as audio files.

HTML is a type of coding language used on web pages to display text, images and audiovisuals. Documents on the internet that are not **encoded** in HTML are not web pages.

USING A WEB BROWSER

A web browser is a type of software that lets you browse websites or web pages. Every time you Google something, you are using a web browser to read what is on the web page. There are different types of web browsers, for example Google, Mozilla Firefox and Internet Explorer.

In Chapter 5, we looked at the basics of connecting to the internet. Remember, you need to have a working internet connection to test out all of these things!

Something to know

People get confused between a web server and a website, for example, if someone is saying "my website is not loading", they actually mean that the web server is not responding, which then results in the website not loading.

HTML BASICS

Just to pique your interest, you can read a bit more about HTML basics by clicking on this QR code:

https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/HTML_basics

BROWSER BASICS

Have a look at this video to understand the basics of the web browser:

https://www.youtube.com/watch?time_continue=195&v=FxirRVJWUTs

For the purpose of this book, we will use the Google Chrome browser. However, you can use whichever browser you want to. Browsers may have a different look or feel, but they all work pretty much in the same way.

To open a web browser, click on the *Start* menu and enter the name of one of the web browsers. You can then click on the icon to open the browser.

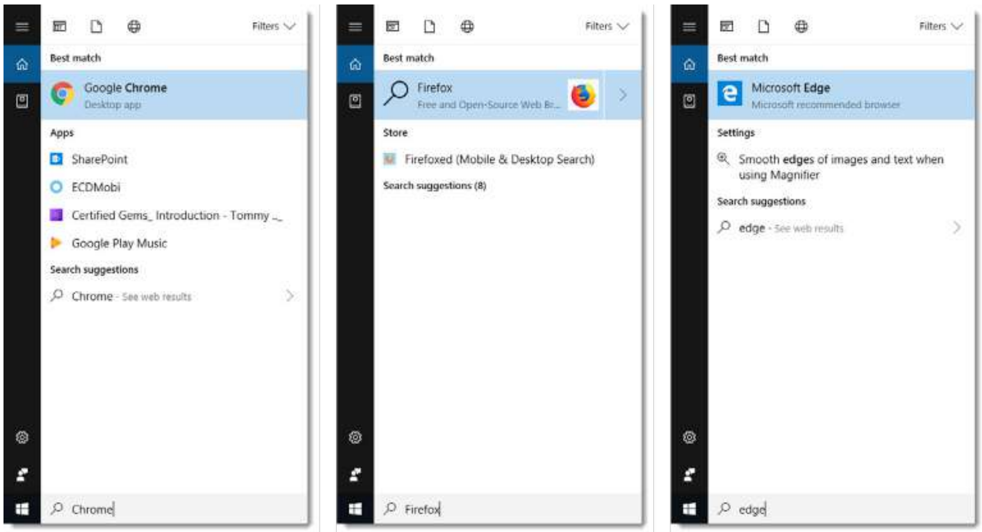


Figure 11.2: *Opening the web browser*

You will learn more about how to use the browser later in this chapter.

To browse to a website, you must know its unique web address, such as www.instagram.com for the Instagram website. Once you know the address, you can enter it into the address bar at the top of the browser.

Guided Activity 11.2

Do the following activity in the class under the guidance of your teacher.

Figure 11.3: *Entering the web address*

As soon as you press *Enter*, the web page should begin to load.

... continued

Guided Activity 11.2

… continued

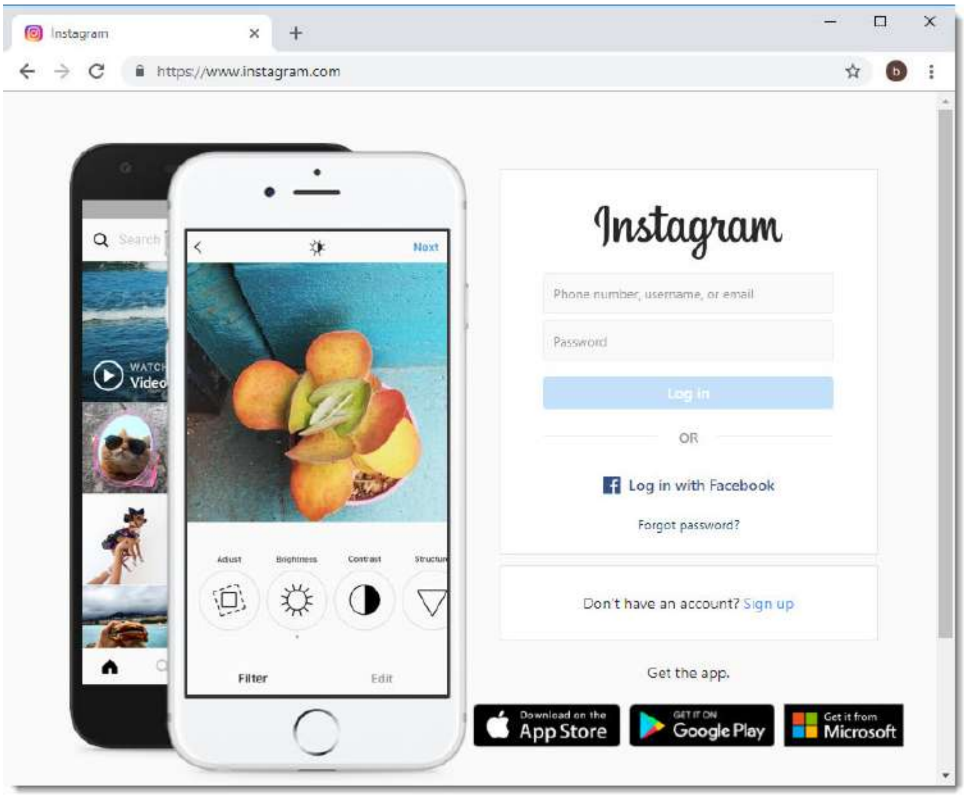


Figure 11.4: *The Instagram web page*

You can now browse the website by clicking on buttons and links, or following the instructions on the page.

WHAT IS THE DIFFERENCE BETWEEN A WEB PAGE AND A WEBSITE?

A web page refers to the text, images, or graphics displayed in a web browser. A web page is generally a single page of content on a website. You can access a web page by entering the URL into the address bar of a web browser.

A website is a collection of web pages that are usually linked by **hyperlinks**. For example, if you go to the *Sunday Times* website, it consists of many web pages. On each page, you will find various articles, columns and content that have been grouped into different categories.

To help you understand the difference between web pages and websites better, look at the example on the following page.

Something to know

If you are still a little bit confused about websites, think of it as a book with many pages.

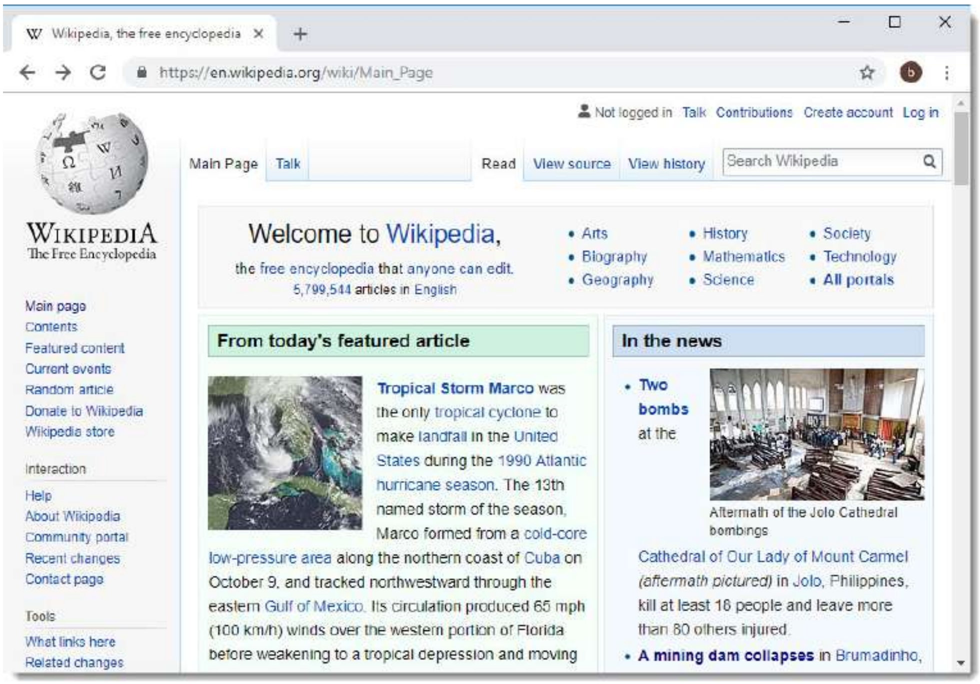


Figure 11.5: *Wikipedia is an example of a website that has millions of web pages*

There are many websites on the WWW; from how to remove a carpet stain, to the latest news on what is happening in the world. However, for the purpose of this book, we will focus on the following types of websites and their purpose.

Table 11.1: *Types of websites*

TYPE	PURPOSE	EXAMPLES
Blog, weblog or vlog	A website that posts short, informal stories about any topic. It is a good place to find information about a topic, place, or a hobby. This is usually run by an individual or a group of people. A vlog is a video blog or video log. It is a blog where all, or most of the content is in video format.	• http://www.cakewrecks.com/ • https://boingboing.net/ • https://mentalfloss.com/ • https://catoverberg.wordpress.com/ Many examples of vlogs can be found on YouTube.
Social network	A website that connects you to people by making friends, seeing what they are up to, and posting your thoughts and photos.	• https://www.facebook.com/ • https://twitter.com/
Web application	An application that runs directly on a website, such as a word-processing application, or a fitness tracker.	• https://docs.google.com/ • https://www.fitocracy.com/
Wiki	A website where people from across the world can edit or modify the information on a website. The most famous is *Wikipedia*, the online encyclopaedia. It also provides links to the original sources so that you can check whether the information is correct or not.	• https://en.wikipedia.org/ • https://tvtropes.org/

Different types of websites

Make a screen capture video with text and voice-overs, explaining the different types of websites. For each type of website, you can quickly explain what the purpose of that type of website is, and then show an example that meets that purpose.

URLS, URL SHORTENER AND THE ADDRESS BAR

Each web page on the internet has its very own unique address called a URL, which tells the internet exactly what page you want to see on a website. Think of a URL as a street address that tells the web browser where to go on the internet.

When you type a URL into the address bar of the web browser and press *Enter*, the browser will take you to that specific page. For example, in the figure below, we typed *www.bbc.com* in the address bar (highlighted in green), which will then load the BBC web page.

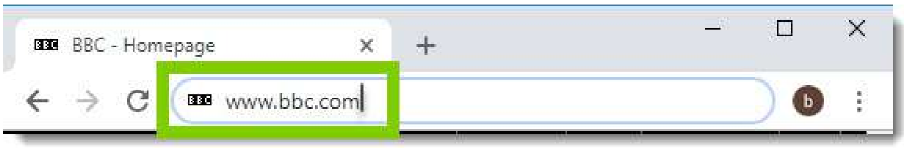


Figure 11.6: *Typing in a URL in the address bar*

In the following example, you will learn more about URLs. Each segment in a URL is a part that makes up the web address.

The ***domain** name* is the most important part of an internet address. This could be a word or a phrase that an internet site has identified as the name of the website. People use it to find information on the internet, for example businesses use it to get people to visit their websites.

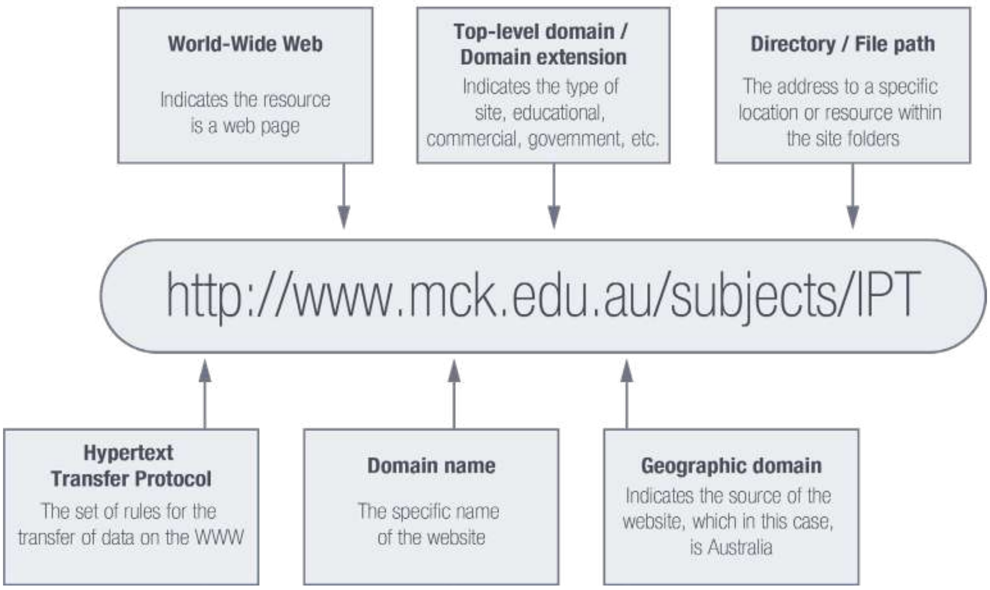


Figure 11.7: *The different parts of a URL*

The tables on the following page show the different codes you could come across indicating an organisation type and the country of origin.

URL SHORTENERS

A URL shortener is an online application that converts a normal URL into a much shorter format. The user has to copy the website address into the URL shortener application and the tool will convert the address to a much shorter one.

UNDERSTANDING URLS

Watch this video to understand the different parts of the URL:

https://www.youtube.com/watch?time_continue=14&v=5Jr-_Za5yQM

Guided Activity 11.3

Do the following activity in the class under the guidance of your teacher.

1. Open the website: https://bit.do/.
2. Copy then paste the following address to shorten it: https://www.blog.google/products/maps/wheres-waldo-find-him-google-maps/
3. Click on "Shorten".
4. What were the results?
   Something like this – http://bit.do/eCuFC?

Each time you do this, a different URL is generated.

Table 11.2: *Codes showing the different types of organisations*

CODE	TYPE OF ORGANISATION
.com	Commercial
.co	Registered company
.ac	Academic institutions
.org	Organisation
.net	Network providers
.edu	Education

Table 11.3: *Examples of country codes found in URLs*

CODE	COUNTRY
.au	Australia
.ca	Canada
.uk	United Kingdom
.us	United States of America
.de	Germany
.za	South Africa

LINKS

When you see a word or phrase on a web page that is blue or underlined in blue, it is usually a hyperlink or link in short. Links are used to navigate the Web. When you click on a link, it will take you to a different web page. Also, sometimes when clicking on a link, the mouse cursor will change to a hand icon before you open the link.

In this example, we have opened a Wikipedia web page and have clicked on the Gauteng link to learn more about it.

Figure 11.8: *Clicking on the link*

Clicking on the Gauteng link will take you to the following web page:

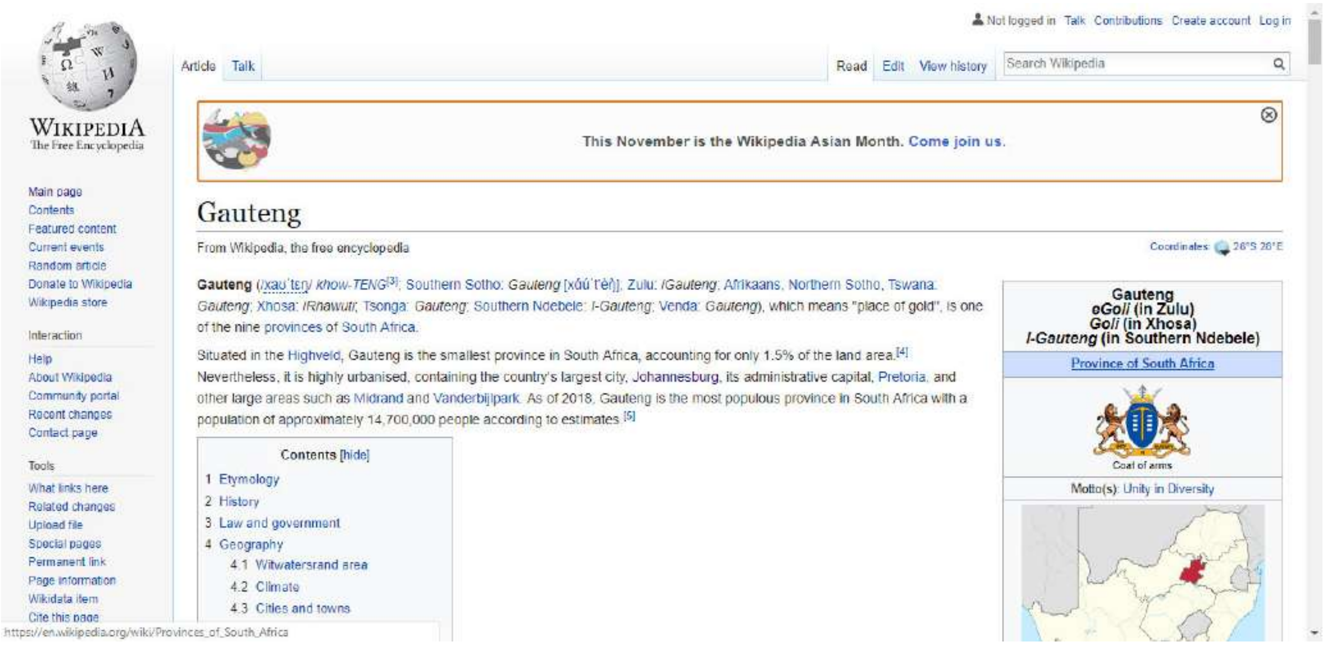


Figure 11.9: *The opened web page*

Each of the blue-coloured words in the text are hyperlinks to other web pages.

Guided Activity 11.4

Do the following activity in the class under the guidance of your teacher.

Many websites use images as links. After clicking on the image, it will take you to a new page. In the example below, if we click on the image, it will open a page with more information about it.

Figure 11.10: *An image link*

Take note that links do not always go to other websites. Links can also allow you to download a file, such as a music file, software file and so on. When you click on a link like this, it will download the file to your device. In the example below, an installation file for a new application can be downloaded.

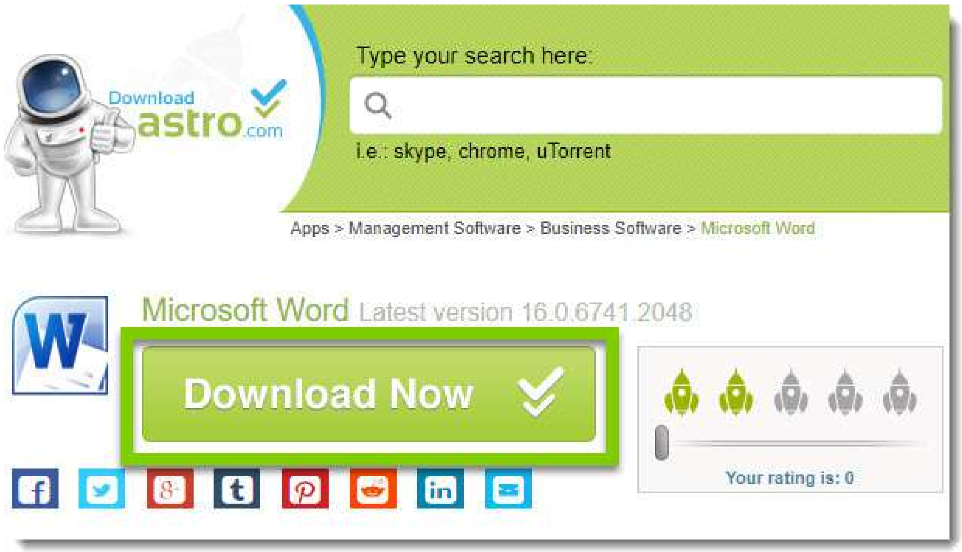


Figure 11.11: *Link to download a file*

As you have learned, links do not all look the same. Links can have the following different forms:

- Text that is blue and underlined
- Images, such as photos
- Tabs on a web page
- Text that is not underlined, but **bolded** and appears in another colour

Links play a very important role when using the Web. It allows you to access different web pages, navigate between these pages and download files.

NAVIGATION BUTTONS

Navigation buttons are found on the web browser and allow you to do many things. The arrow buttons, which are called the *back* and *forward* buttons, let you go to the websites that you have recently visited. If you click and hold on one of the buttons, you can view the recent browsing history.

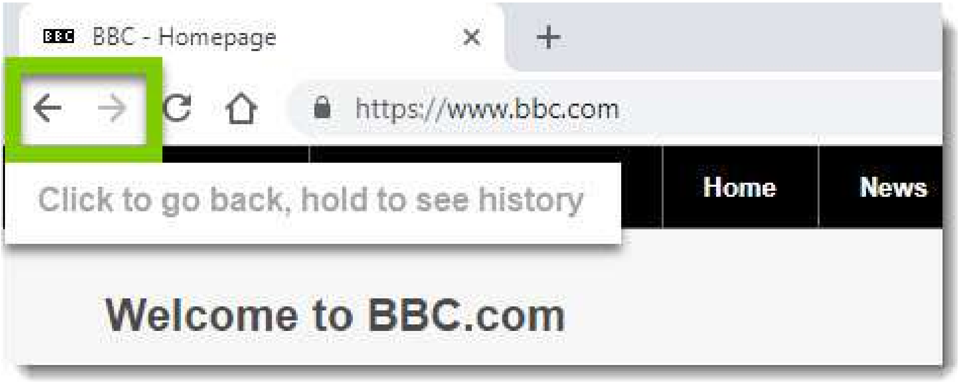


Figure 11.12: *Back and forward buttons*

The *Refresh* button will reload the current web page that you are on. If a website stops working, you can just click on the *Refresh* button.

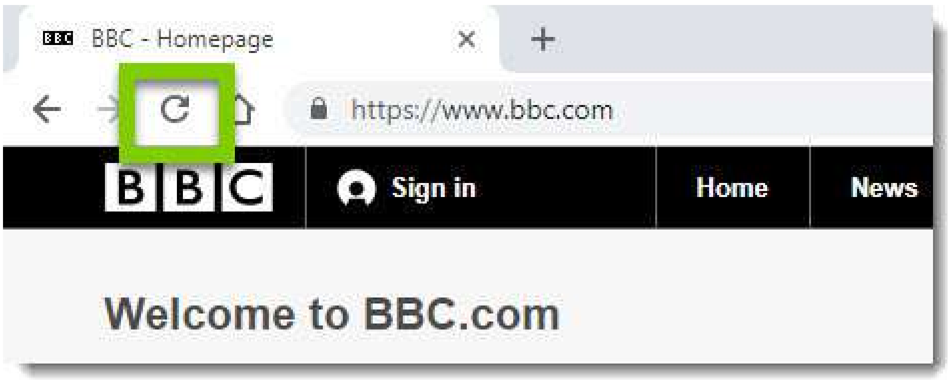


Figure 11.13: *Refresh button*

TABBED BROWSING

Many web browsers let you open links in a new tab, which is great because you can open as many links as you need and they will all stay in the same browser window. This prevents separate browser windows from being open on the computer screen.

Guided Activity 11.5

Do the following activity in the class under the guidance of your teacher.

To open a new tab, you can do the following:

1. Right click on the link and select *Open link in new tab.* Keep in mind that the wording might differ from browser to browser.

2. To switch tabs, click on any tab that is not selected.

3. The web page will then open.

4. To close a tab, click on the X.

… continued

Guided Activity 11.5 *… continued*

5. To create a new blank tab, click on the button found on the right of the open tabs.

ADVANTAGES OF TABBED BROWSING

Some of the advantages of using tabbed browsing, include the following:

- Allows the user to view many web pages at once.
- If users come across a link in the web page they are currently viewing, they can click on the click and a new web page will open in another tab without closing the current web page.
- Allows the user to move between web pages without actually closing any web pages.

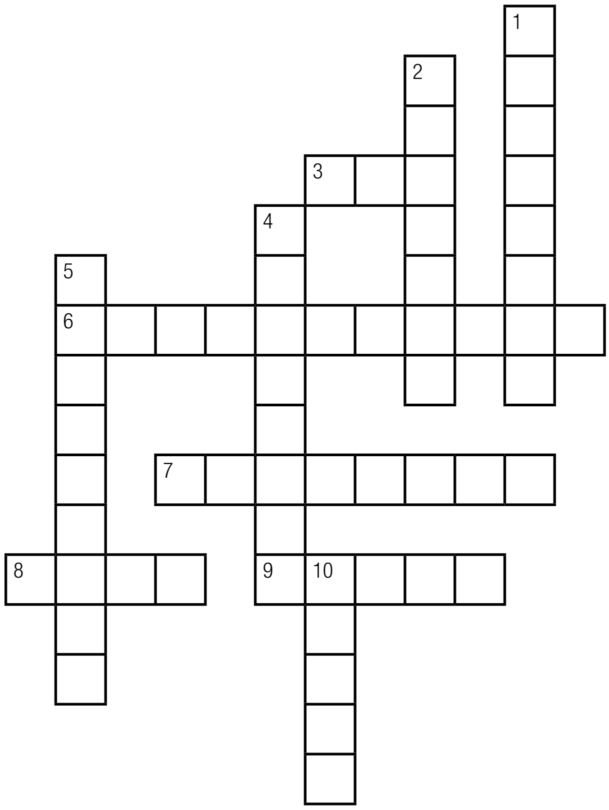
Take note

You can use the following useful hotkeys when browsing. Open the web browser and try doing the following using the keyboard:

- **Open a tab:** *Ctrl+T*
- **Switch between tabs:** *Ctrl+Tab*
- **Close a tab:** *Ctrl+W*

Activity 11.1

1. Briefly explain the difference between a website and a web page.
2. Complete the crossword puzzle below by answering the questions that follow:



Down:

1. A social network website
2. A page made up of text, pictures, audio clips, video clips and animations
4. A website about a single person
5. An example of a news website
10. Brings several types of websites together

Across:

3. When you open a link, it opens in a new ___________
6. A website you can check load-shedding schedules
7. A website that posts short and informal stories
8. A website that collaborators can edit and modify
9. An example of an educational website

UNIT

11.3 Search engines

In the world we live in today, more and more things are being done online. You need basic computer skills to do research, be social and do many other things on the internet. The ability to search for information on the internet is an important skill to have and by improving this skill, you can find what you are searching for without going through many irrelevant websites.

Since there are billions of websites on the Web, there is a lot of information available. Search engines make access to this information much easier. We will look at the basics of using a search engine and some basic techniques on how to get more useful search results.

SEARCHING FOR INFORMATION

Browsing to the correct website usually only works well when you know the address of the website, or if you need to visit a specific website. However, it is not practical to keep a list of web addresses on you all the time. You might be looking for something specific on the internet, shopping for something new but need to compare prices, or you might just be looking for new interesting websites. For that, you need a **search engine** to find the information you are looking for.

Three popular search engines are:
1. Google (www.google.com)
2. Microsoft Bing (www.bing.com)
3. Yahoo (www.yahoo.com)

Of the three search engines, Google is the most powerful and easy-to-use search engine.

BASIC BROWSING AND SEARCHING TECHNIQUES

In the following activity, we will search for "grey matter".

Guided Activity 11.6

Do the following activity in the class under the guidance of your teacher.

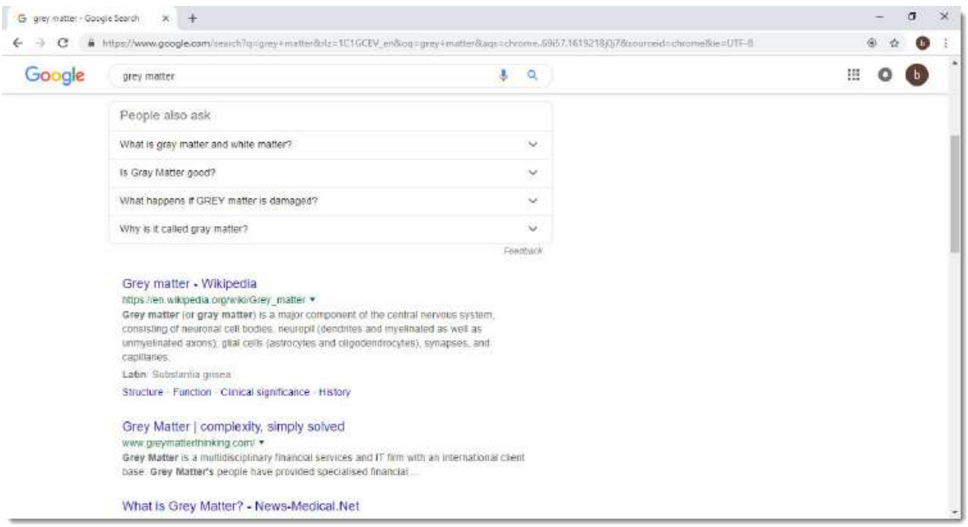
To search for something on the internet, you can do the following:
1. Open the web browser and navigate to a search engine. Most web browsers generally allow you to search directly from the address bar. However, some browsers might actually have a separate search bar next to the address bar.
2. Type in one or more keywords (this is also known as the *search term*).
3. Press *Enter* on the keyboard.
4. After you have pressed *Enter*, you will see a list of all the relevant websites that match the search words. If a site looks interesting or looks like what you are looking for, click on the link to open it.

… continued

Guided Activity 11.6

… *continued*

5. **Skim** through it.



6. If it is not exactly what you are looking for, return to the results and look at other websites.

BASIC SEARCH STRATEGIES

Have a look at this video to get a better idea of search strategies:

https://www.youtube.com/watch?v=7RIB1CJovTs

REFINING YOUR SEARCH

If you still have problems finding the exact website, you can use the following special characters to help refine your search:

- To exclude a word from the search, type in a hyphen (-) at the beginning of the word. For example, if you want to find grey matter results without mention of the movie with the same name, you could search for *grey matter – movie*.
- At the same time, if you use a (+) before the beginning of the word or phrase, the search results will show results with just the movie mentioned e.g. *grey matter + movie*.
- You can also search for the exact words or phrases, which can give you much better results. To do this, use quotation marks (“ ”) before and after the search words. For example, if you search for “grey matter songs”, the search results will only have songs about grey matter.

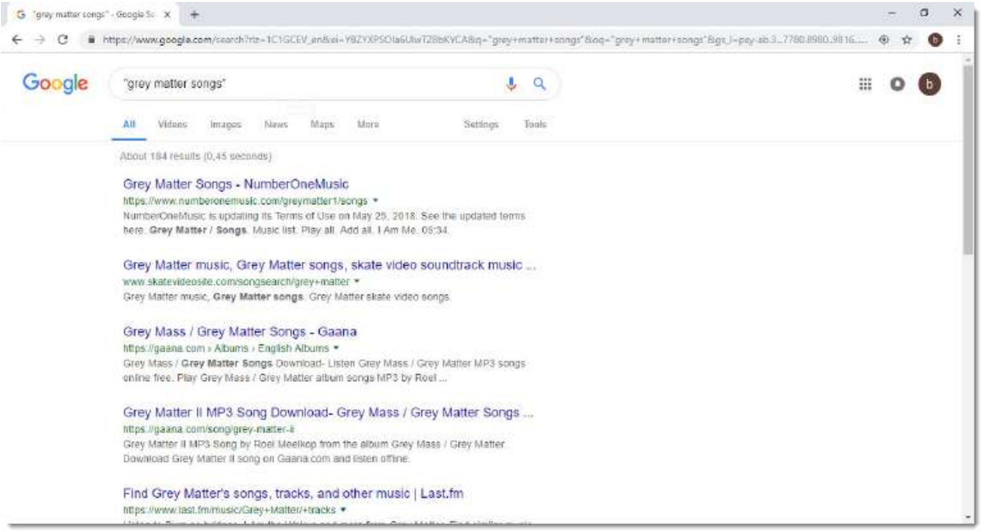


Figure 11.14: *A refined search*

The best results are usually shown on the top page, with the least popular or visited results appearing on the second and third search pages. Since the first search results are usually the best, if you do not find what you are looking for on the first page or two of the results, you should try a different search phrase, or try to use an advanced search technique.

SEARCH TECHNIQUES

The following table highlights a few of the most useful search techniques.

Table 11.4: *Search techniques*

PURPOSE	METHOD	EXAMPLE
Search for a specific type of web page	Once you have opened Google Search, click on the *Images, News, Videos* or *Maps* button to search for those items.	Search: *Meghan Markle* Click on *News* and then type in *Meghan Markle.*
Search for web pages updated at a specific time	Once you have opened Google Search, click on the *Tools* button. Click on *Any time* and then select the time period for which you are looking.	Search: *Meghan Markle* • Click on *News* • Click on *Tools* • Click on *Any time* • Select *Past week*
Search for results from a specific website	Add the phrase *Site*: followed by the website to the search query.	Search: "violin music" site: youtube.com
Search on social media	Add the name of the social media website after the @ symbol in the search query.	Search: "violin" @twitter

These techniques can be very useful. However, because search engines have become extremely efficient, you can even find information without using these "special" techniques.

Guided Activity 11.7

Using the information that you just learned, do the following activity during class. Your teacher will guide you through this activity.

1. Open a web browser.
2. Go to the Wikipedia website by typing in its URL: *https://www.wikipedia.org/.*
3. Click on the *English* tag.
4. Search for *Bafana Bafana.*
5. Click on the hyperlink *Nickname*. What does the double use of the name Bafana Bafana mean?
6. Open a new tab.
7. Type in *South African rugby.*
8. Click on *News*
9. Click on *Rugby: South African schools with the most Springboks*. According to this news article, which school has the most Springboks? How many does it have?
10. Right click on the tag *Golden Lions* at the bottom of the page.
11. Click *Open link in new tab*. What does the heading say about the Super Rugby final?
12. Open a new tab.
13. Search for *Sunspots without the inclusion of climate*.
14. To what do the search results you found on the first page refer?
15. Search for *Sunspots with the inclusion of climate.*
16. What do you find in the search results? List the names of the first three websites.
17. Do you see any reference to sunspots on the skin? Why or why not?
18. Close all the websites by clicking on the X of each tag.

By now, you might have come across the terms downloading and **uploading**. Downloading is when the computer or smart device receives a file or data from the internet. Uploading, on the other hand, is when the computer or smart device sends a file or data to somewhere on the internet.

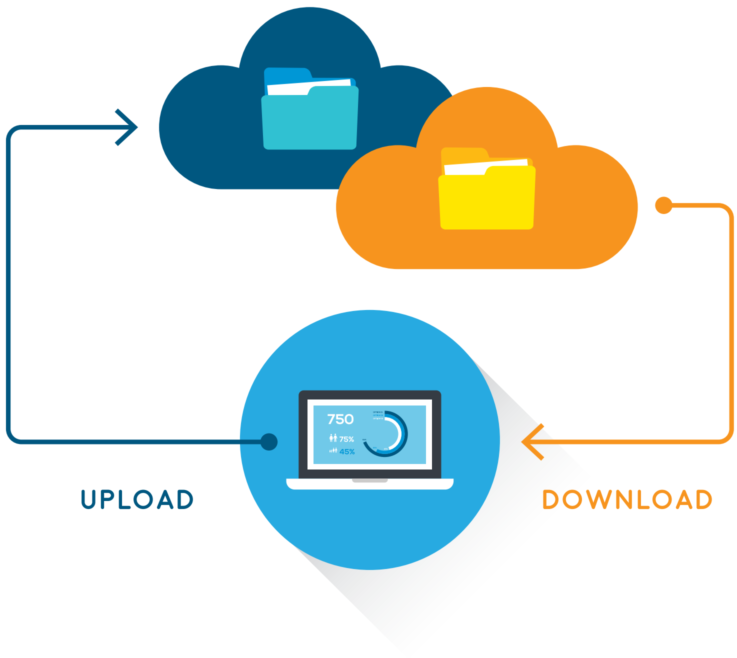


Figure 11.15: *Uploading and downloading with a PC*

You have probably, at some point, downloaded or uploaded a file or data from the internet. Do you remember downloading a music file or posting (uploading) a photo on your Facebook wall or Instagram?

DOWNLOADING

To be able to download something from the internet, it is an important skill to learn as you will use it throughout your lifetime to download software programs, music, photos, documents, videos, etc.

To take you through a download process, you can look at the following example of downloading a video-player software program.

Guided Activity 11.8

Do the following activity in the class under the guidance of your teacher.

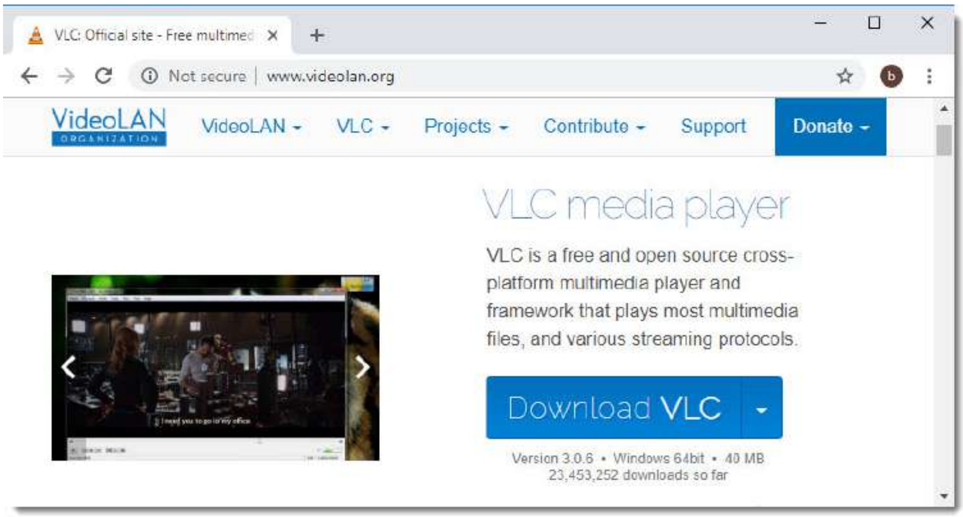
To download VLC, you can do the following:

1. Use the web browser to go to the website for VLC: *http://www.videolan.org/.*
2. Once the website has loaded, find and click on the *Download VLC* button. This will download the approximately 40 Mb-VLC installer onto the computer.

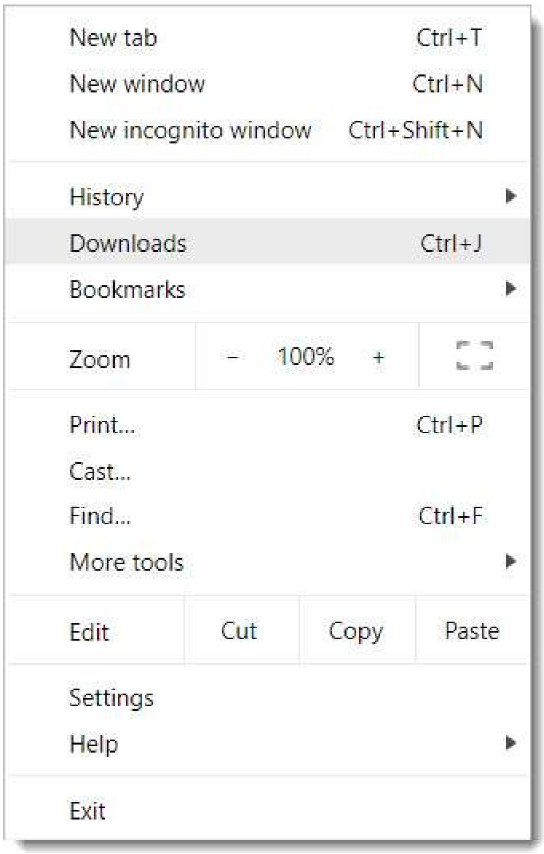
… continued

Guided Activity 11.8

… continued



3. When the download has completed, the file will be saved on the computer, or it will open with the program that you selected.
4. If you have a problem finding the download, you can always do the following:
   Click on the three vertical dots found on the top right-hand side of the screen and choose *Downloads* from the drop-down menu. You will be able to see all the files that you downloaded and from that, you can choose the file you need.



Something to know

Some browsers or even files do not have the "automatic" download process when you click the link to the file. In such a case, you will have to right click the link and choose the *Save link as* option, and then choose a location to which you can download the file.

Guided Activity 11.9

Do the following activity in the class under the guidance of your teacher.

1. Open the web browser.
2. Go to Google web page.
3. Google the following: Types of input devices "doc".
4. From the search results, browse for the website that best describes input devices.
5. Click on the website name.
6. Follow the instructions to download the document.
7. Open the document in Word.
8. Save the document as a PDF and call it: *Uploadeddoc*.

UPLOADING

You can use the upload function to send emails, post photos on a social media site, upload files to Google Drive, etc.

If a site enables uploads, it will have an upload option to help perform the upload or file transfer. Every site has a different uploading process. A dialogue box opens after you click on the *Upload* button. For example, Facebook has an icon that is an image. This is Facebook's *Upload* button for an image, starting the upload process. You can upload an image or file to a website that allows it.

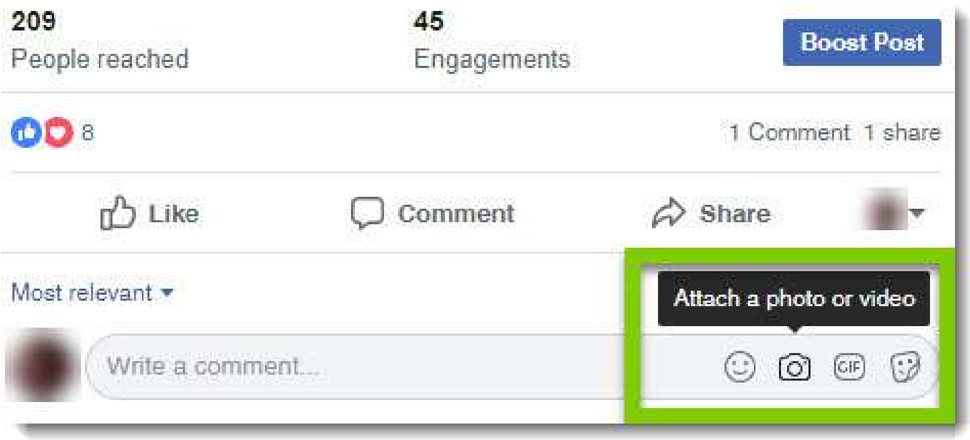


Figure 11.16: *Upload button on Facebook*

After clicking on this button, the dialogue box will appear, prompting you to select a file. Navigate to the location to where the file is stored and click on the *Open* button. A progress bar will track the upload process.

Some websites have a drag-and-drop interface. For example, when logged into Google Drive, you can upload files or even zipped folders from the computer to the browser window by dragging and dropping the file.

Let's look at the following example of *Ted Talks*, an organisation that uploads videos from expert speakers about a variety of topics, such as education and business.

Guided Activity 11.10

Do the following activity in the class under the guidance of your teacher.

To upload a video on a website e.g. *Ted Talks*, you can do the following:

1. First of all, the user needs to log into the TED Media Uploader. If this is your first time, you can request the uploader here.
2. Once you have logged in, select the event category of the video that the user wants to upload.
3. Click on "Upload Video" to upload a new video.
4. Follow the instructions, for example the type of file you are loading, as well as other personal details.
5. Include a video description.
6. Click "Upload".

Activity 11.2

1. Which one of the following is NOT a good technique used to refine an Internet/web search?
   A. Using one word only
   B. Specifying the domain
   C. Using quotation marks
   D. Using operators such as "and" and "not"
2. Define the following terms:
   a. Search engine
   b. Web application
3. Complete the table below by indicating in Column B whether the action in Column A is downloading or uploading. Only write the number and the answer down.

COLUMN A	COLUMN B
Putting photos up on a Facebook wall	3.1
Using a torrent to legally download movies	3.2
Getting music from beemusicplayer	3.3
Getting a PDF document from the internet	3.4

REVISION ACTIVITY

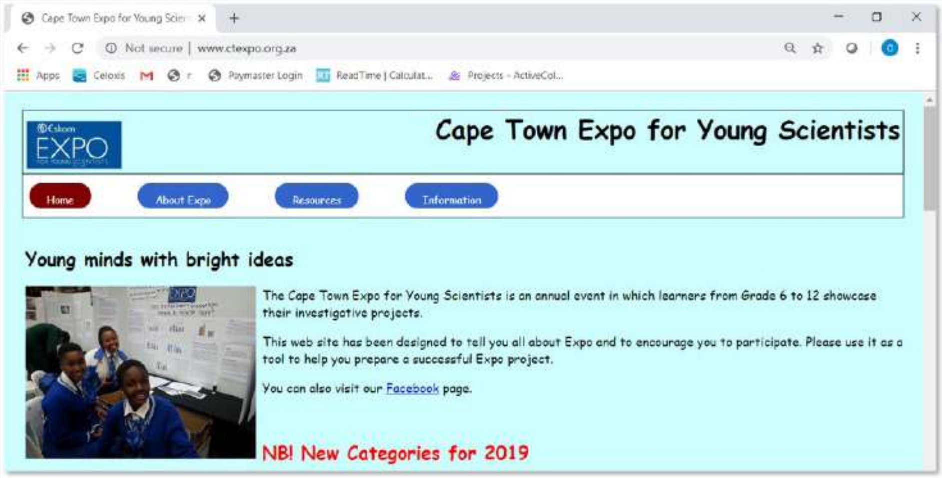
1. Write down the letter that matches the correct answer:
   The letters "http" stand for:
   A. Hypertype transport protocol
   B. Hypertext transfer protocol
   C. Hypertext transfer process
   D. Hypertype transfer protocol (1)
2. Which of the following web browsers is installed with Windows 10?
   A. Google Chrome
   B. Mozilla Firefox
   C. Edge
   D. Safari (1)
3. Your friend thinks that the internet and the WWW are the same thing. Explain to her what the difference is between them. (4)

… *continued*

REVISION ACTIVITY

... continued

4. Study the screenshot below that shows the home page of a website and use it to answer the questions that follow:



a. What is the URL of the web page? (1)
b. What is the domain name of this web page? (1)
c. What is the geographical domain of this web page? Which country is this? (2)
d. Name two other web pages that are part of this website. (2)
e. Why is the word "Facebook" in blue and underlined? (1)

5. For each of the scenarios below, write down the type of website. Choose your answer from blog, weblog, social network, wiki or web application.
   a. Zanele has a website where she keeps a journal of the exotic foods she enjoys (1)
   b. Rebogile has created a website that people can use to create a monthly budget and to track their expenses (1)
   c. Julia, a Life Sciences teacher, has a website where her learners can go to find study tips, as well as answers to recent examination questions (1)

6. The screenshot below shows part of the Google Chrome browser. Use it to answer the questions that follow:

a. The Google Chrome browser uses tabbed browsing. Mention two advantages of using tabbed browsing. (2)
b. Which tab is currently active? Explain how you arrived at your answer. (2)
c. What is the function of the + button after the second tab? (1)
d. What is the meaning of the closed padlock in front of the URL in the address bar? (1)
e. Give the name of another browser other than Google Chrome. (1)

... continued

REVISION ACTIVITY

… continued

7. Francois wants to take some friends that are visiting from Europe to the top of Table Mountain in Cape Town. He plans to use the cableway to travel to the top of the mountain. Francois uses the Google search engine and types in "Cape Town" in the search box. He gets about 650 000 000 results.
   a. How could Francois change his search phrase to only get results for the Table Mountain cableway? (2)
   b. Name another search engine that Francois could use, other than Google. (1)
8. Explain the difference between downloading and uploading when using the internet. Give an example of each. (4)

TOTAL: [30]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

NO.	DID YOU …	YES	NO
1.	Understand the different concepts used with the Web.		
2.	Understand the difference between a web page and a website.		
3.	Learn about the different types of websites and their purpose.		
4.	Identify the different parts of a URL and what they mean.		
5.	Understand what tabbed browsing is and how it is useful.		
6.	Learn the different techniques on how to use search engines.		
7.	Understand the concept of uploading and downloading.		

TERM 3

INTERNET COMMUNICATION

CHAPTER 12

CHAPTER OVERVIEW	
Unit 12.1	Electronic communication devices
Unit 12.2	Email as a form of e-communication
Unit 12.3	Basic emailing

At the end of this chapter, you should be able to:

- Describe what electronic communication is.
- Describe the different electronic communication devices.
- Electronically communicate using a PC.
- Discuss the different electronic communications.
- Explain the difference between ISP versus web-based email.
- List and explain the different features of email, such as the “Cc” and “Bcc” fields, attachments, and address books.
- Compose email messages.
- Understand how to send, receive, forward, Reply to and Reply to all using email.
- Apply netiquette rules over email.
- Understand email etiquette.

INTRODUCTION

In the past, communication between people was done either face to face, using the telephone, or by writing letters. Now, we live in a world where electronic communication or better known as e-communication has taken over. Think about it, when last did you go to a family or friend gathering and no one took their phones out to message a person? This is how much electronic communication has taken over.

Electronic communication refers to any data, information, words, photos, **emojis** and symbols that are sent electronically to one or more people. This can be done through emails, social media, newsgroups, chat rooms, video conferencing, instant messaging, phone and fax. With just one click on the send button, you can say “hi” within seconds to your friend in Spain!

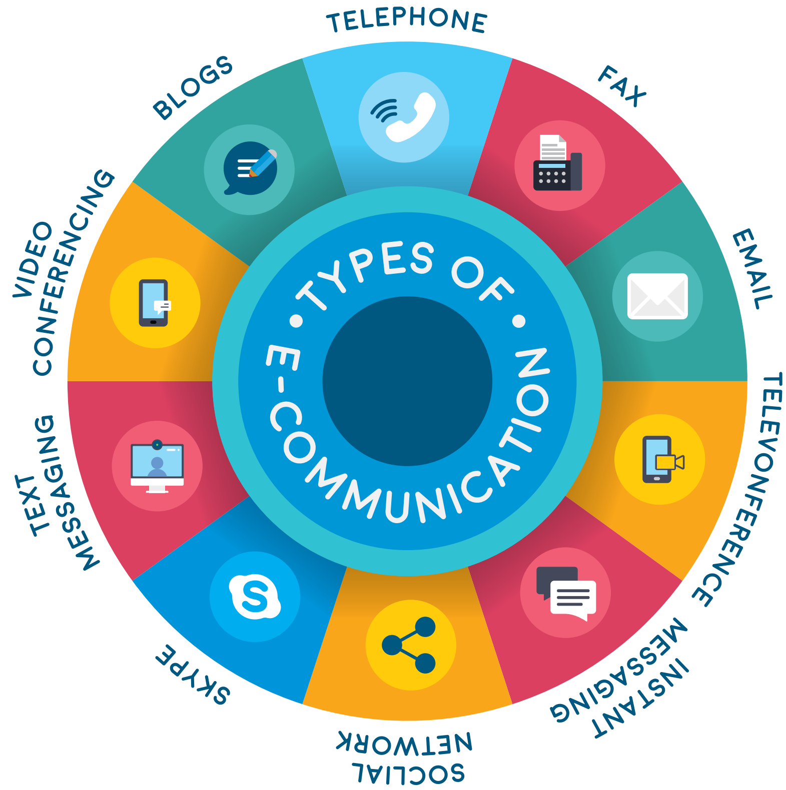


Figure 12.1: *Examples of e-communication*

Up to a few years ago, email was the best way to communicate with people online. However, these days there are many other ways to communicate, allowing you to do things such as:

- Making phone calls using your computer
- Sharing the same message with many people at the same time without sending the same message individually
- Interacting with different platforms on the internet and making comments and statuses, or even sending messages
- Seeing the person on the other side of the world you are talking to through video conferencing

In this chapter, we will look at the different electronic **communication devices** and the different types of applications used in electronic communications. We will also look at email and how to compose and send basic email messages, as well as basic email **netiquette**.

UNIT

12.1 Electronic communication devices

An electronic communication device refers to any type of computerised device (instrument, equipment, or machine) with software that can compose, read, or send any electronic message using radio, optical or other electromagnetic systems. An electronic message can be a text message, email, an instant message such as WhatsApp, teleconferencing, social networking, Skype, blogs, or even access to an internet site. It can consist of signs, signals, writings, images and sounds, or artificial intelligence (AI).

When people are not communicating online, they make use of phone calls, face-to-face conversations and written letters, depending on each situation. The same applies to online communication; people make use of instant messaging, social networking or email, depending on the situation. You can choose the mode of communication that best suits you.

The following table shows how different people communicate online. We asked Zanele, Linda and Vusi how they communicate online. Their answers were as follows:

Table 12.1: *How people communicate online*

 Zanele	 Linda	 Vusi
I use instant messaging to chat to my friends and family. I use email to communicate to my teacher about assignments and questions I have.	I love my blog, I use it to communicate my ideas and hobbies to the rest of the world. I also use video calling to call my parents every week.	I use email to chat to my clients and other co-workers. I use text and picture messaging to communicate to my cousins in the USA.

In this chapter, we will look at e-communication using a PC, with specific focus on using and receiving emails.

E-COMMUNICATION USING A PC OR MOBILE DEVICE

There are many ways in which to use electronic communication when you have an internet connection. The most common type would be electronic mail, which is usually referred to as email, but there are also other forms, such as video chat, discussion forums, Skype and instant messaging. In this section, we will look at the different applications that are used to facilitate electronic communication.

EMAIL

One of the first and most popular forms of electronic communication is email, which allows users to send messages and files over the internet. One of the best things about email is that you do not have to wait for weeks to receive it. This type of mail arrives moments after it has been sent.

Something to know

Have you ever tried to think how many emails are sent per day around the world? In 2017, an average of 269 billion emails were sent and received each day. In 2018, 124.5 billion business emails and 111.1 billion consumer emails were sent and received each day.

Something to know

According to recent statistics, teenagers regard email as a more formal mode of communication and they usually use it for school or exchanging messages with adults. Teenagers find emails to be too slow and time consuming, compared to instant messaging.

Email is used in the following ways:

- Communicating with clients or with other employees
- Keeping in touch with friends and family
- Sending files as attachments to another person
- Sending marketing messages to potential clients

We will learn more about emails and how to send them a bit later in this chapter.

MAILING LISTS

Jenny receives an email from Takealot.com every day before 05:00 with their latest sale items and items that they have on promotion. Do you think that someone is sitting in front of a computer sending emails at 03:00 to everyone that subscribed to this website? That is definitely not the case; they use a mailing list.

A *mailing list* is a collection of names and addresses used by an individual or organisation to send information or materials to multiple recipients. There are two main types of mailing lists:

1. **Response list:** This list consists of names and addresses of people who have responded to an offer of some kind. Because these people are known responders, their names are generally priced higher than those in lists compiled by other means.
2. **Compiled lists:** These lists contain the names and addresses of people obtained from telephone directories, public records, direct mail and telemarketing campaigns, etc. This information is sometimes also obtained from a list broker who researches, analyses and evaluates the many individual lists available.

Other types of mailing lists include the following:

- **Team list:** Mailing lists that teammates use to communicate with each other within their teams
- **Group list:** Mailing lists used by a specified group of people to communicate with each other, for example a list of the parents of each learner in the school
- **Event list:** Mailing lists communicating information around a specific event that is happening, such as the upcoming athletics inter-schools event

MOST POPULAR INSTANT MESSAGING APPLICATIONS

Watch the following video:

https://www.dw.com/en/the-worlds-most-popular-instant-messaging-apps/av-40684764

INSTANT MESSAGING

The internet has changed the way in which we communicate, with email being the most popular form of electronic communication. However, sometimes even email is just not fast enough. Let's look at the following example to understand this better.

INSTANT MESSAGES

In the smartphone-obsessed world in which we live, more people are sending messages through web-based application, such as WhatsApp, WeChat, Slack, LiveChat, Hangouts, Lync, Telegram and Snapchat.

These applications are short messages that are sent and read in real time, allowing you to communicate much quicker than normal emails.

There are also browser-based type instant messaging applications that do not require downloading, for example Facebook and Gmail. These applications allow you to chat to your contacts whenever you are logged in.

Chat and instant messaging are mainly used when both, or all of the people are online so that everyone can read your message instantly, hence instant messaging! An email, on the other hand, will not be seen until the recipient actually checks the email, making instant messaging much more efficient for quick messaging.

Instant messaging is an online chat that allows you to exchange text messages, symbols, pictures and even documents, in real time, over the internet. It also allows you to see if a friend or co-worker is online and apart from sending text messages and files, you can also enjoy video and voice chats.

Instant messaging usually includes a list of your contacts (called a "Buddy List") which lets you see who is online. This type of messaging is best used when it is one-on-one communication, but it is possible to send several people messages at the same time.

WEB BROWSERS

There are different types of web browsers available, such as Google Chrome, Mozilla Firefox and Internet Explorer. Without these web browsers, it will be impossible to view web pages and websites! In the past, users had to download the software application to their PC so that they can chat, listen to music and watch videos. Nowadays, all these things can be done by just opening the web browser. All you need to do is open a web browser to access a website so that you can communicate via social networks, forums, emails and popular instant messaging services.

Websites are also designed to communicate with the user; whether it is a shopping site that lets their visitors know they have a 70%-off sale, or a small blog where one person talks about the different recipes they try out.

TEXT AND PICTURE MESSAGING

Text and picture messaging are very popular and are usually sent from one smartphone to another. There are, however, applications that allow you to text and send pictures from your computer to a smartphone, or the other way around. Text and picture messages are fast and only take seconds to reach the other person. It is also useful to text someone if it is not possible to call the person, for example if you are in a meeting or in class.

It is easy to send text, pictures or even voice messages through different applications, such as WhatsApp, SnapChat, Facebook, Hangouts and many more.

People can instantly send these forms of multimedia anywhere in the world and it does not cost much.

VIDEO MESSAGING

Video messaging is an easy way to make inexpensive phone calls anywhere in the world and from your computer. It does not even require a very fast internet connection. Many instant messaging and chat services have voice chat and allow you to talk to friends who are online. Video chat lets you see and hear friends, family or clients in real time. Whether you are talking to a friend or someone at work, video chat can add a personal touch to your chats!

MOST POPULAR ACRONYMS AND ABBREVIATIONS

To check the most popular acronyms and abbreviations when texting, click on this code:

https://www.netlingo.com/top50/popular-text-terms.php

Something to know

To decrease the cost of making voice calls, internet companies invented technologies that allow you to make voice calls over the internet (VOIP). Services, such as WhatsApp and Skype, allow you to make free voice calls to other WhatsApp and Skype users.

WEBLOG

Have you ever wanted to create your own website, but never figured out how to? Well you can and best of all it is free and quite easy. Today, it is possible to create a website by creating a blog (short for weblog). So, what is a blog?

A blog is an online diary or journal that is located on a website and presented with the newest information first. It can contain text, pictures, videos, animated GIFs and even scans from old physical offline diaries or journals and other hard-copy documents. Although it is mostly run by individuals or small groups, it can also be run by an organisation, promoting itself and its products or services.

FAX TO EMAIL

A fax (short for *facsimile*) is an exact copy of a document (text or images) that was scanned and transmitted as data by a telecommunication link. This usually goes to a telephone number that is associated with a printer, or other output device.

Fax-to-email, or email fax, is a system that allows users to send or receive a fax using email. This form of communication is used when you use the internet to send faxes. Basically, what happens is that the sender faxes the document using the receiver's fax-to-email number, who will then receive the fax as an attachment in his or her email inbox.

Activity 12.1

1. Explain the term "e-communication".
2. Describe an e-communication device and list three examples.
3. What do you understand by the term "mailing list"?
4. What application do you use the most to text and send pictures? How do you use text messaging?
5. Who normally uses blogs?
6. Vusi has no fax machine. However, his friend from Brazil must fax him a document that he urgently needs. His friend only has access to a fax machine.
   a. What can Vusi do to make sure that he gets the document?
   b. How do you think Vusi should go about doing it?

12.2 Email as a form of e-communication

Email is an electronic form of communication that is exchanged between people through computers, or other electronic devices, such as smartphones or tablets. You need the internet to send emails.

You also need software in the form of an email application that allows you to send, receive, forward and reply to email messages. With emails, you can also attach files, such as documents, photographs and even videos (restricted to a certain data cap).

In this section, you will learn more about email, how email addresses are written, and the features and tools that are included in having an email account.

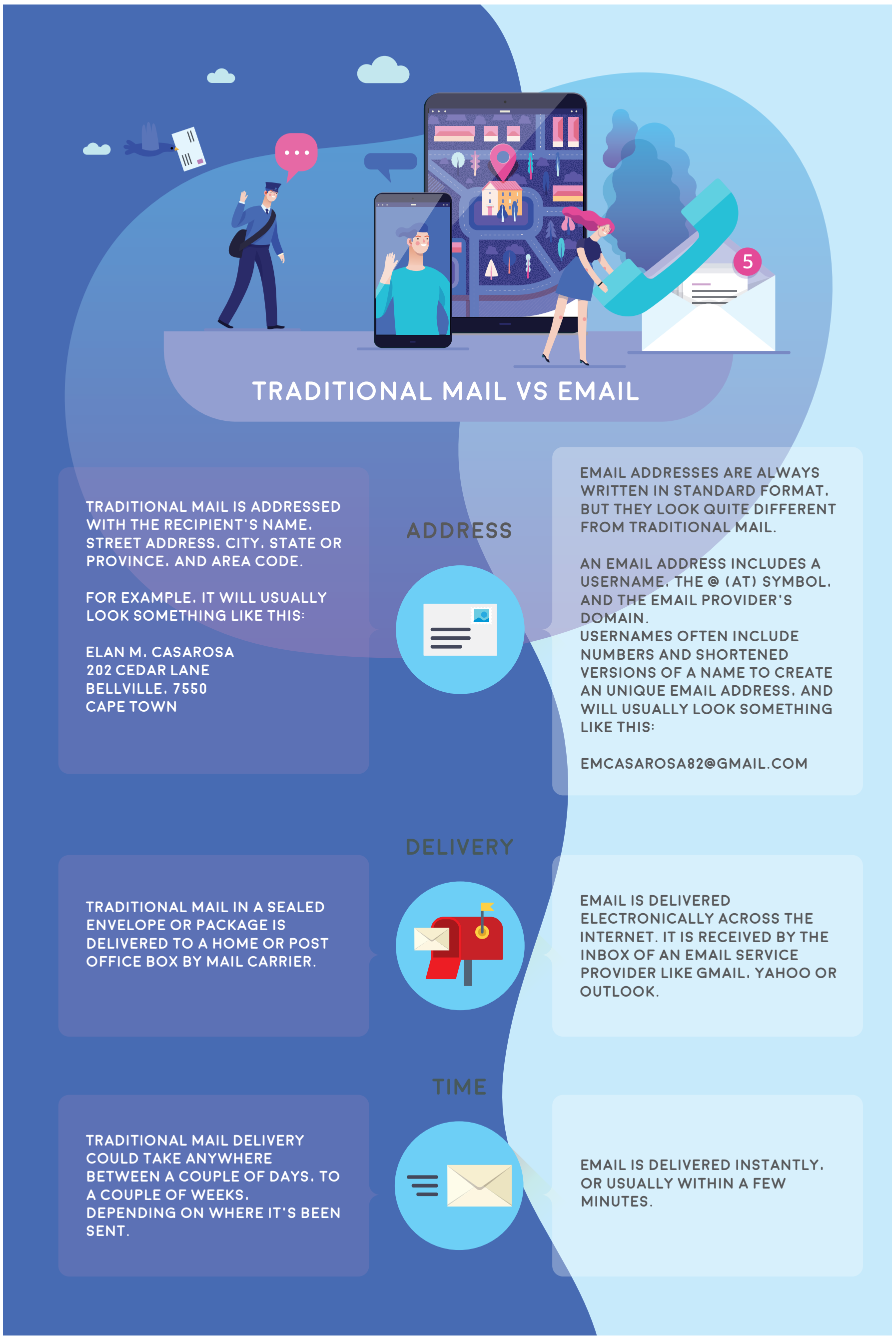


Figure 12.2: *Traditional mail versus email*

Something to know

You know that email is a way to send and receive messages using the internet. It is similar to traditional post, but with some important differences. The figure on the left shows what email is about, as well as its advantages.

COMPONENTS OF AN EMAIL ADDRESS

To receive emails, you will need an email account and an email address. To send emails to other people, you will need their email address details too. An email address is, therefore, the unique identifier for an email account.

It is important to understand how to type an email address correctly because, if entered incorrectly, it might not be delivered to the recipient, or it might even go to the wrong person!

Email addresses are always written in a standard format that consists of two parts, a local part or *username* and a *domain*-part, separated by an @ symbol. The local part is used by the receiving mail server to determine where the email must go and what must be done with it after it arrived at its destination.

These different parts are called an email's **taxonomy**. The taxonomy of the email address, computer@mweb.co.za, is explained below:

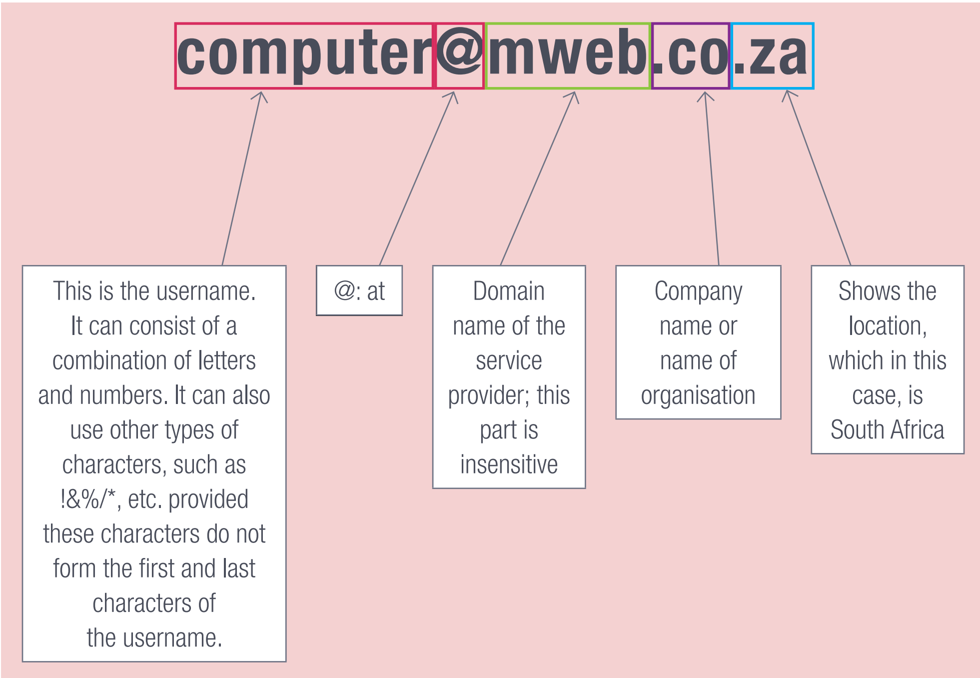


Figure 12.3: *Components of an email address*

ISP VERSUS WEBMAIL

The first thing to decide when you are looking for an email service provider is whether to use a **webmail**, or to turn to an ISP to take care of your emails. As both have their advantages and limitations, it boils down to how much money you are willing to spend and the importance of maintaining your email.

Table 12.2: *ISP versus webmail*

	ISP	WEBMAIL
Advantages	The main advantage of having an ISP email account, is the support that you get when something goes wrong.	• Webmail does not charge for its services. • Webmail refers to any email service that you can reach through a web browser. It also means that you can check for email messages on any computer with a web browser installed; whether you are at home, at work, or on holiday. • You can also keep the same email address, even if you change your ISP.
Disadvantages	• An ISP provides a mailbox to end users as part of their paid services. When using an ISP, your emails will be on the ISP's servers. This means that you will have to connect to the ISP mail server to download your emails. If anything happens to these servers, there is no way that you can get your emails until the problem is fixed. • If you have the wrong incoming or outgoing password, you will not be able to send or receive emails. • ISPs charge for their services. • Should you move or change your ISP, you will most probably have to get a new email address. • There may be a limit to the amount of storage space for emails and attachments.	The main disadvantage is that it will send advertisements to your Inbox to help cover its costs. Some services, such as Gmail, will look for keywords in your email messages and show you relevant advertisements.

Top webmail providers currently are Google's Gmail and Microsoft's Outlook.com. They are the most commonly used as they allow you to access your email account from anywhere in the world, provided that you have an internet connection. You can also access webmail from a smartphone or tablet!

HOW EMAIL WORKS

The following happens when an email is sent:

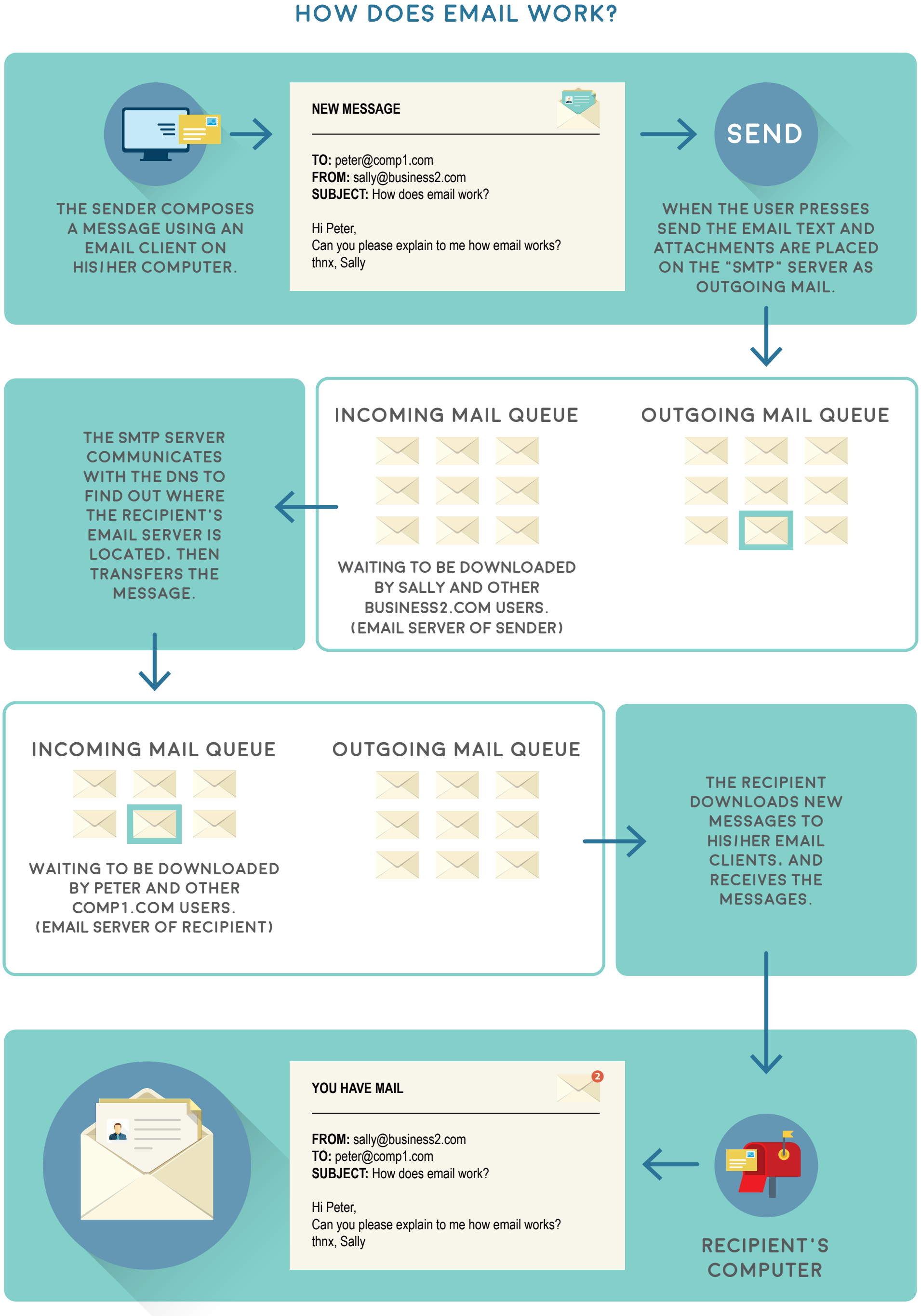


Figure 12.4: *How email works*

Activity 12.2

1. Which of the following emails in the tables are valid or invalid? Explain why it is invalid each time.

EMAIL ADDRESS	VALID OR INVALID?	WHAT MAKES IT INVALID?
a. @domainsample.com		
b. anonymous@domainsample.com		
c. –autumn-dancer@domain.com		
d. jane.doe43@domainsample.co.uk		
e. janwa kitty@domain.com		
f. janwa_kitty@domain.com		
g. joesmith.nospamplease@nospam.example.com		
h. john.doe@.net		
i. john.doe@domainsample.ne		
j. john.doe43@domainsample		

2. What is the difference between an ISP email and a webmail?
3. Give two advantages and disadvantages of ISP and webmail.

Before composing and sending email messages, you first need to know how to use an email account. In this section, we will use Gmail to show you the basics of using an email account. If you decide to choose another email provider, the interface might be different, but the basics will still be the same. You will learn how to:

- Sign up for an email account
- Navigate and familiarise yourself with the user interface
- Compose, respond, receive and forward email messages

In this section, we will look at the email interface, the terms and actions used in email, as well as features commonly used with email.

EMAIL INTERFACE

Whichever email service provider you choose, you will still need to get to know the email interface. This includes the *Inbox, Message* pane and *Compose* pane. Although the interfaces will look different depending on the email provider, in the end, they all function in the same way.

INBOX

The *Inbox* is where you will see and manage any emails that you receive. Emails are listed according to the date or time received, the name of the sender and the subject of the email message.

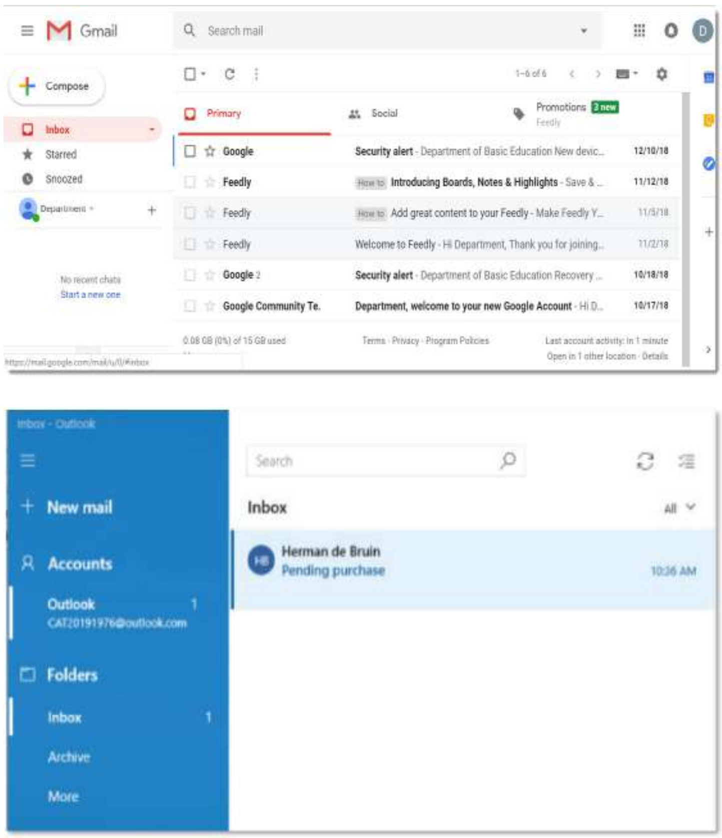


Figure 12.5: *An example of an email interface*

MESSAGE PANE

After you select an email in the *Inbox*, it will open the *Message* pane where you can read it and then choose how to respond, using a variety of commands.

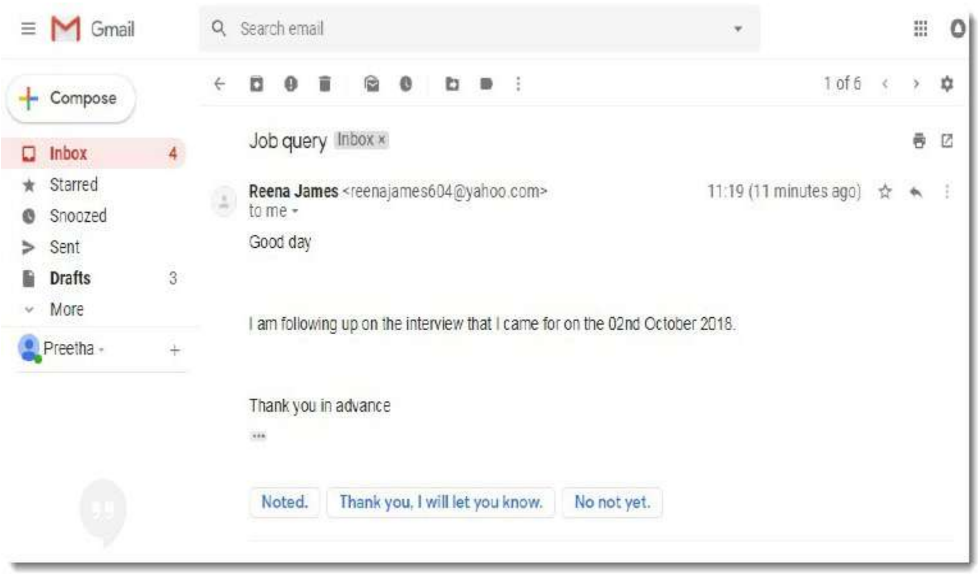


Figure 12.6: *The Message pane*

COMPOSE PANE

When you click on the *Compose* button from the *Inbox*, it will let you create your own email message. From here, you will have to enter the recipient's email address and a subject. If you need to upload files, such as photos or documents, you can do that by adding an attachment.

Figure 12.7 shows the different parts of the *Compose* window.

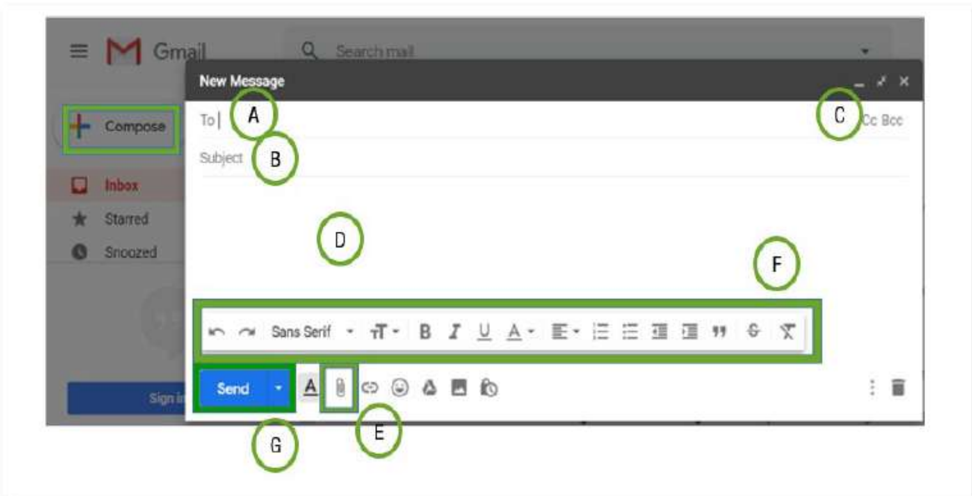


Figure 12.7: *Compose a New message window*

A: Recipients are the people to whom you are sending the email. For each recipient, you will have to type in the email address. In most cases, you will add recipients to the "To: Field", but you can also add recipients to the "Cc": or "Bcc": fields.

B: **Cc** stands for carbon copy and is used when you need to send an email to a person who is not the main recipient. It helps to keep everyone updated, but at the same time, it lets the person know that he or she does not need to respond to the email. **Bcc** stands for blind carbon copy. It is almost the same as Cc, but the email addresses in the Bcc fields are always hidden. This type of emailing is perfect if you need to send the same email to a large group of people, but keep their email addresses private.

C: The subject of the email is used to say what the email is about. The subject should be short, but clearly state what the message is about.

D: This is the body of the email. The body of the email is the actual text of the email, similar to that of a normal letter. It starts with a greeting, adds a paragraph or two and ends with a closing statement with your name at the end of it.

E: An attachment is a file, for example an image or document, that can be sent along with the email message by clicking on the *Attachment* button. Gmail allows you to attach more than one file, as long as it is not bigger than 25 Mb in total. However, by integrating its cloud storage service, Google Drive with Gmail, it is now possible to attach files as large as 10 Gb.

F: The *Formatting* button allows you to access the different formatting options for example, changing the font colour, look and size of the message, as well as include hyperlinks.

G: When you are happy with the message, you can click *Send* to send it to the recipient(s).

Something to know

If the details of person that you are sending an email to is already in your address book, then you can actually start by typing in the person's first name and Gmail will display the contacts below the "To:" field. Then press *Enter* to add the person as a recipient. This saves you time from actually typing in the whole address.

SENDING EMAILS

Now that you have created an email account and understand how the email interface works, you can start sending email messages.

In this section, we will look at how to compose an email, add an attachment, reply and forward emails.

When you type an email, you will be using the *Compose* window. In this window, you will add the email address of the recipient(s), subject and the message itself. You will also be able to add one or more attachments in this window.

HOW TO SEND AN EMAIL

Watch this video to understand how to send an email:

https://youtu.be/2eHOJbEE-6k

Guided Activity 12.1

Do this activity in your class. Your teacher can help guide you in creating an email account. Use the account you created for this activity.

To send an email, you can do the following:

1. Click on the *Compose* button found in the left-hand pane.
2. The *Compose* window will appear on the right-hand side of the page.
3. In the "To" field, add one or more recipients. This is done by either typing in the email addresses manually, or by using the *Address Book*.
4. Type a subject for the email message.
5. Type a message in the body field. When done click *Send*.

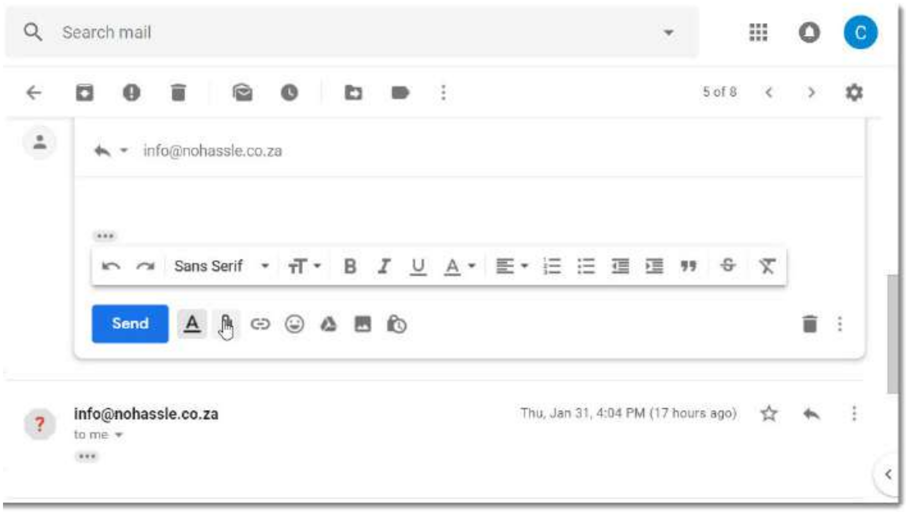
ATTACHMENTS

An attachment is a file, image, or document that is sent with the email. For example, if you are applying for a bursary, you can send your results and other documents as an attachment, while the body of the email serves as the covering letter. It is always a good idea to let the recipient know that you have attached a file or files.

Guided Activity 12.2

Let's look at how to add an attachment. Do the following activity in your class with the help of your teacher.

1. When you are writing the email, click on the paperclip (attachment icon) icon at the bottom of the *Compose* window.



12.8: *Dialogue box with the attachment icon*

A dialogue box will open with files that you can choose to attach.

2. Choose the file you want to attach and click *Open*.

12.9: *The Open window*

... continued

Something to know

Always remember to attach the file before sending it, it is extremely common for people to send an email without actually attaching the file!

Guided Activity 12.2 *… continued*

The attachments will then start to upload. Most attachments just take seconds to upload. Large attachments can take longer.

Figure 12.10: *Upload status bar*

3. When you are ready to send the email, just click *Send*.

You can click on the *Send* button before the attachment is finished uploading.

REPLYING TO EMAILS

Emails are not only sent; they are received as well. After reading an email, there are certain actions you can take, for example, replying to the message, forwarding the email to someone else, or opening an attachment if there is one.

READING EMAILS

Any email that you receive will be in the *Inbox* and you can tell which emails are unread, because they are marked in bold. From the *Inbox*, you can see the name of the sender, the subject of the email, as well as the first few words of the email. This means that, before you even open an email, you can already gauge a few things about it.

Guided Activity 12.3

Do the following activity in your class with the help of your teacher. This activity will help you learn how to read an email:

1. Go to the *Inbox* and click on the email you want to read.

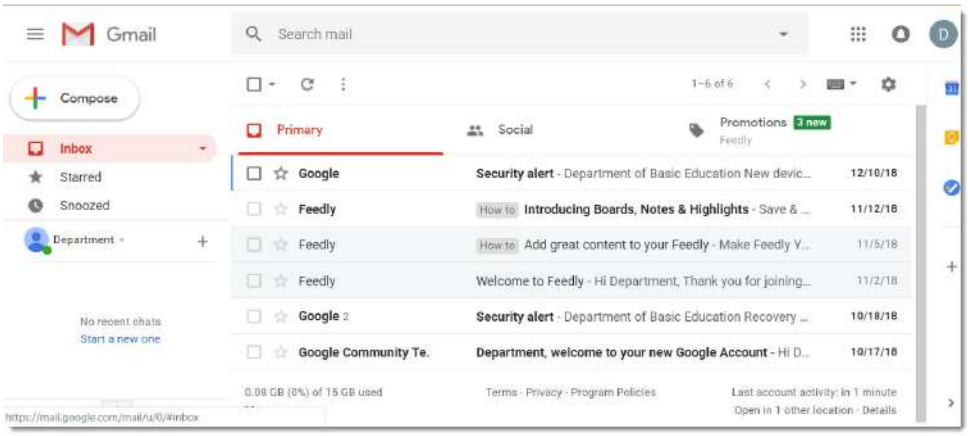


Figure 12.11: *Inbox*

… continued

Guided Activity 12.3

… continued

The email will then open up in the same window:

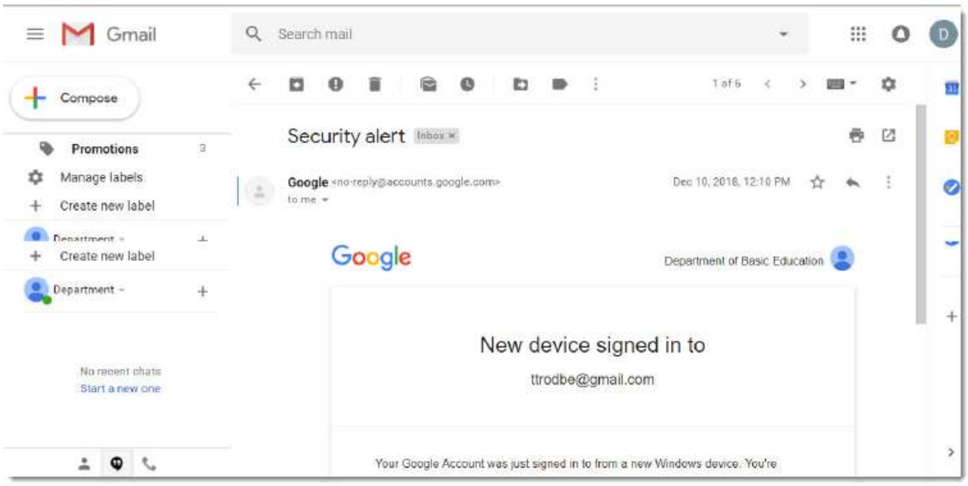


Figure 12.12: *Opened email message*

After reading the email, you can choose to *Reply*, *Reply to all,* or *Forward* the email to someone else. To perform any one of these actions, click on the three-dot icon found on the right-hand side of the email window.

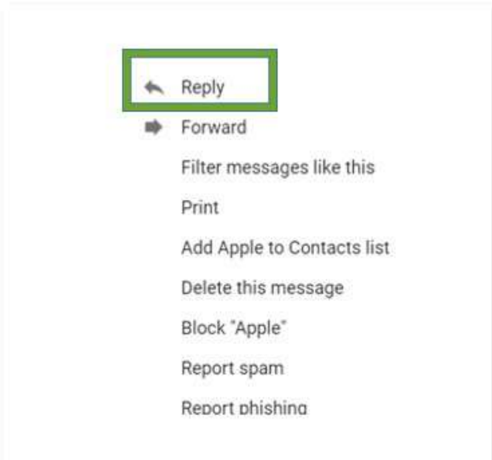


Figure 12.13: *Reply, Forward and Reply to all*

REPLY, REPLY TO ALL AND FORWARD AN EMAIL

What is the difference between *Reply*, *Reply to all* and *Forward*?

Table 12.3: *The difference between Reply, Reply to all and Forward in an email*

TYPE	DESCRIPTION
Reply	When you click on *Reply*, this sends a response to the person that sent you the email. So, even if people were Cc'd in on the email, they will not receive the response.

… continued

DEALING WITH EMAILS

To understand more about the different actions you can take after reading an email, watch the following video:

https://youtu.be/a9e7XNo4agE

Something to know

You should only open attachments from a trusted source. Some attachments could contain a virus, especially if it ends with the *.exe* in the file name.

HOW TO ADD A HYPERLINK

Have a look at these videos on how to add a hyperlink in Outlook and Yahoo Mail:

https://www.youtube.com/watch?v=OFIFZ6-xuJc

https://www.youtube.com/watch?v=3iqO0W_-uc0

TYPE	DESCRIPTION
Reply to all	When you click on *Reply to all,* this sends the response to everyone that was a recipient, including those that were Cc'd in the original email. Take note that, if anyone was Bcc'd in the original email, they will not receive the response, even if you click *Reply to all.*
Forward	By clicking on *Forward*, this will send the message to one or more people (group) and will include any of the attachments that were initially included in the original email. The person or people to whom the email was forwarded, will see all the details that were in the original email. You can also edit the original message and remove any attachments before forwarding the email.

ATTACHMENTS

There are times when you will receive an email containing attachments. To view the attachment, you will need to download it. In some cases, for example if the attachment is a Word document or an image, you can view it in the web browser.

In the *Inbox*, even before opening the email, you can tell if the email has an attachment by looking for a paperclip icon to the right of the subject.

Figure 12.14: *Attachment button*

HYPERLINKS IN EMAILS

Hyperlinks are added to emails so that the recipient can follow a link to visit a particular website or web page. Many businesses send emails with hyperlinks to promote or market their services or products. Note that the concept of how to create a hyperlink might be different in the various email applications.

Guided Activity 12.4

Do this activity in your class with the help of your teacher. This activity will look at how to create a hyperlink in Gmail.

1. Sign into your Gmail account and compose an email to create a new message, or you can reply to an email that is already in the inbox.
2. Click on the *Link* icon in the Gmail toolbar, which is usually found next to the paperclip icon.

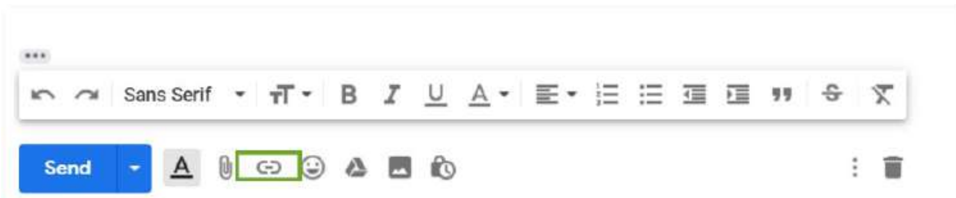


Figure 12.15: *Hyperlink icon*

… *continued*

… continued

3. After clicking on the hyperlink icon, a dialogue box will appear asking you to insert the *Text to Display* and the *Web address*.

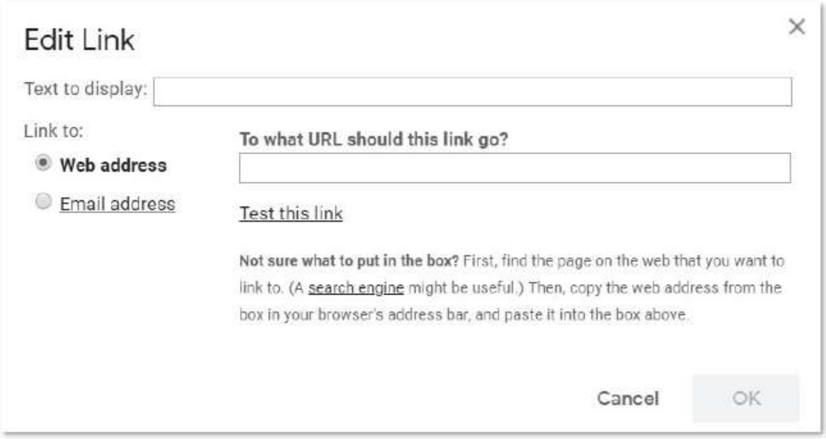


Figure 12.16: *Dialogue box*

4. Enter the web address link you want the recipient to open in the *To which URL should this link refer?* field, such as "www.cnn.com".

Then enter the text you want the hyperlink to appear as in the *Text to display* field, such as *News*.

Then click *OK*.

Figure 12.17: *Adding a link*

The link will then be added in the email, usually in a different colour and underlined so that the recipient knows that it is a link. The recipient can then click on the link and the link will open in another tab.

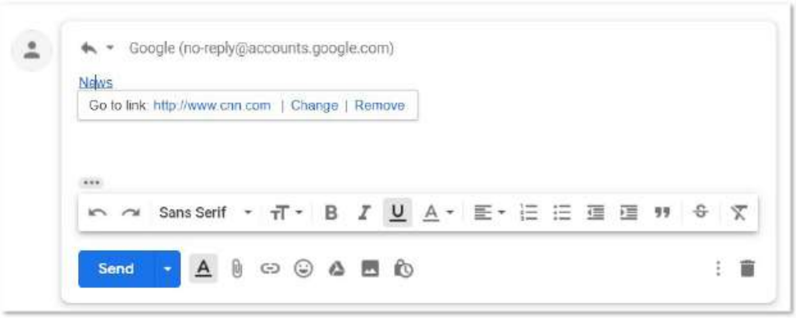


Figure 12.18: *Link added*

Something to know

By pressing *Ctrl+K* in Gmail, the hyperlink dialogue box will open immediately.

ETIQUETTE IN EMAILS

As with any other form of communication, it is always important to practise good **netiquette** in emails. *Netiquette* is short for network and email etiquette. Netiquette means to use good manners when communicating electronically, or when using the internet; whether it is in the workplace, or on a personal level. It is also about respecting other people's privacy.

In this section, we will look at email etiquette, as well as why spelling is important when sending emails. Look at the following guidelines to practise good netiquette:

- **Messages:** Do not **spam** people at work, your friends or family with unwanted email messages or chain emails. Spam refers to unwanted or irrelevant messages that are sent over the internet or through emails.
- **Concise:** Make sure that when writing emails, they are clear and to the point. Also make sure that they do not contain spelling and grammatical errors.
- **Subject line:** Make sure the subject line is clear so that the recipient knows what the email is about.
- **Identify yourself:** Always say who and what you are at the beginning of the email and add a signature with your phone number at the end of the email.
- **Action required:** Let the recipient know right away if any action is required from his or her side. You can do this by marking emails that do not require any action with "FYI" in the subject line.
- **Capital letters:** Do not type emails in capital letters as it gives the recipient the idea that you are shouting.
- **Exclamation marks:** Avoid using exclamation marks or if really needed, use them sparingly. The use of exclamation marks sends a message to the recipient that you are demanding.
- **Large attachments:** Compress large files before sending them. This helps the recipient to save time instead of waiting for a long time to download files. You can always ZIP or compress files (as learned in Chapter 5) to make it easier to send.
- **Gossip, inflammatory remarks and criticism:** Avoid gossiping about others through email, especially at the workplace. Also, do not send insulting, abusive or threatening emails. You cannot withdraw such an email and it can easily be forwarded to unintended recipients. This could lead to unnecessary disputes and grudges in the workplace and in a personal environment.
- **Focus on what is in the email:** Make sure that you read the email properly and address the sender's questions.
- **Proofread the text:** Before sending the email, read through it again to make sure that it is saying what you want it to say and that there are no spelling and grammar mistakes.

Case study 12.1

Julie works at a big auditing firm and deals with different clients on a day-to-day basis. She has made good friends with people at work, so occasionally she gets funny emails from them. However, she found one email to be extremely funny and decided to send it to her friend who works in another company. She just added the friend's name and after a few clicks, the email was sent.

In the space of five minutes, she got some out-of-office replies. She did not realise that she had sent it to all the staff in her department and worst of all, she sent it to her senior manager.

… continued

Case study 12.1

This is an example of the email she sent.

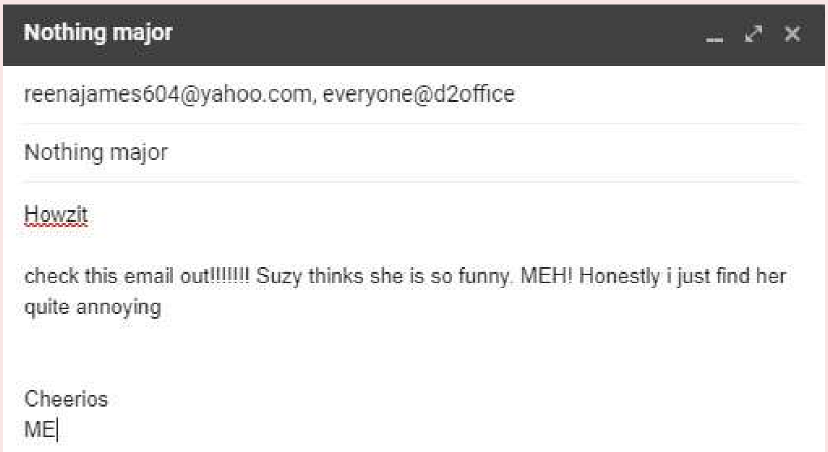


Figure 12.19: *Julie's email that was accidentally sent to everyone in the office*

1. What is the first problem with this email?
2. Discuss two email etiquette rules that are broken by Julie.
3. Explain what Julie can do to save her reputation.

SPELLING AND EMAILS

Spelling is important in emails; whether it is emailing your manager, a co-worker, teacher, or even a friend. You need to make sure that you are sending an email that has no spelling errors. Sending an email with spelling errors does not reflect well on you.

Did you know that your email application has a *Spell Check* tool that is used to correct spelling mistakes when writing an email? In some applications, if a word is spelt incorrectly, it will be underlined in red or green wavy lines. By right clicking on the misspelt word, you can choose the corrected word from the *Context* menu.

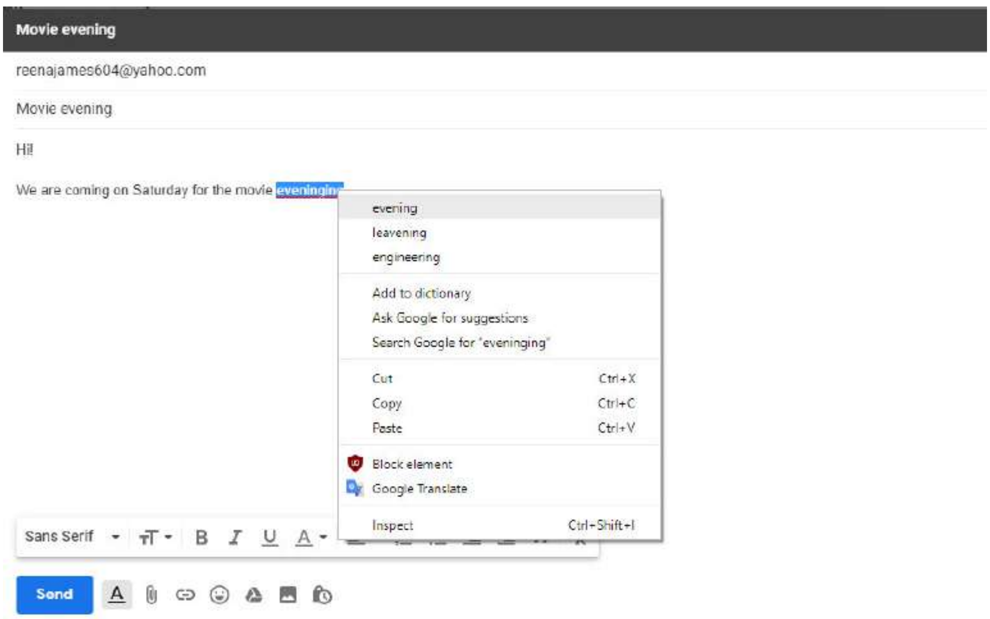


Figure 12.20: *Spell checks*

Something to know

According to Emily Gorton, HR assistant at Powder Byrne Travel Agency, it is a cardinal sin getting someone's name wrong in an email. It shows a lack of interest and no attention to detail, and can negatively change a person's impression about the sender. The best thing to do then is to own up and apologise by sending another email as soon as possible.

It is always important to read the content and grammar of an email before sending it off. Something as small as making an error in someone's name, can make a very big difference in the way people perceive you.

In some email applications, you can enable the *Spell Check* function. This will automatically correct any spelling or grammatical errors.

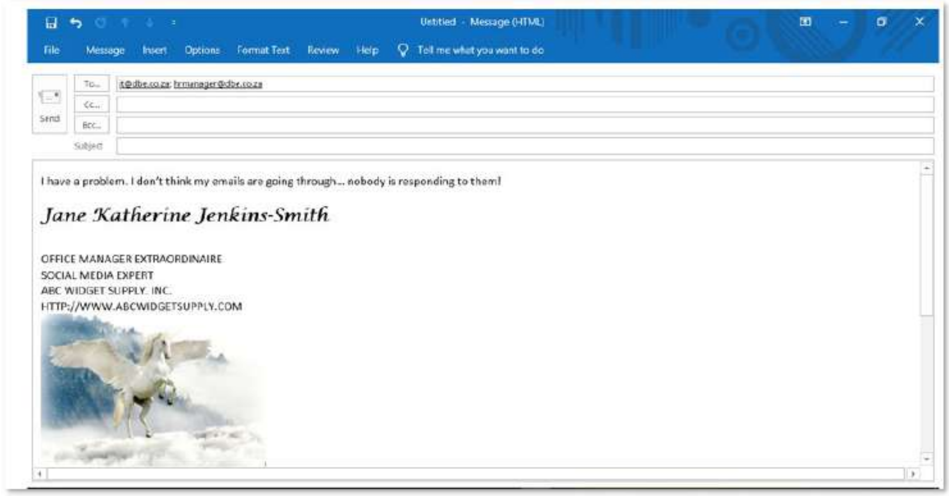
Activity 12.3

1. When sending a file as an attachment via an email, the email bounces back and displays the following message:

> This is the mail delivery agent at messagelabs.com.
>
> I was unable to deliver your message to the following addresses:
>
> Sales@telkomsa.net
>
> Reason: 552 Message size exceeds maximum permitted

   a. Why did the error message occur?
   b. Briefly describe how the error can be fixed.

2. Study the following email:



   a. Compare the [Cc...] and the [Bcc...] section in the email.
   b. Consider the context of the email message on page 196 and list two email etiquette (netiquette) rules that have been violated.

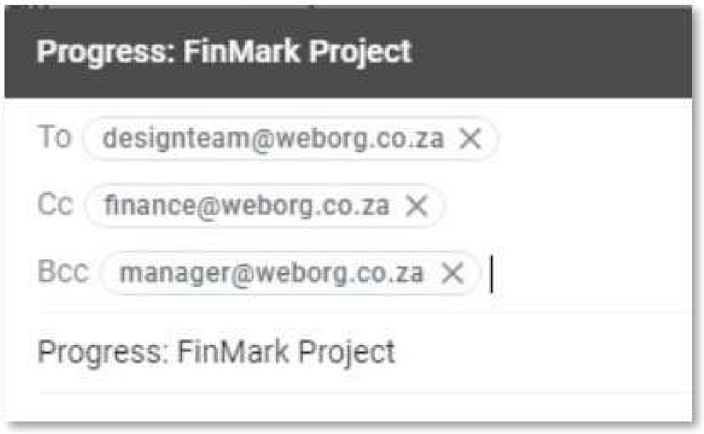
3. Read the following scenario and answer the questions that follow:

> The Motaung family's ISP is *Polkadot*. As part of their contract, the Motaung family receives one free email address. Should they need more email addresses, they need to pay an additional fee per month for each additional email address. Mrs Motaung's business's name is *Haybo! Catering*.

   a. Suggest a suitable email address for Mrs Motaung's business. Motivate why you suggested this email address.
   b. Suggest how Mr Motuang could get his own email address at no cost.

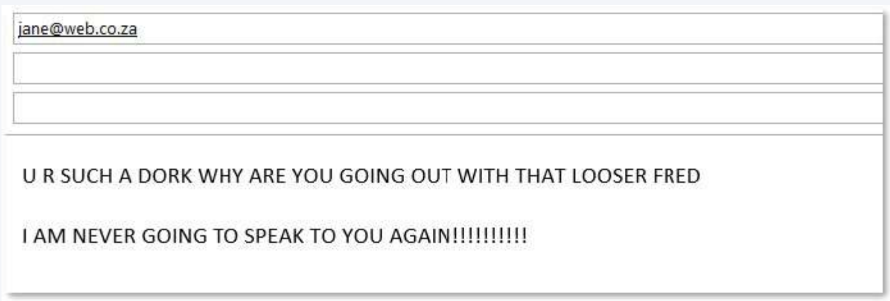
REVISION ACTIVITY

1. Explain what is meant by "e-communication". (3)
2. Write a short paragraph to explain the difference between email and instant messaging. (4)
3. Study the following screenshot and use it to answer the questions that follow:



   a. What is the username of the email address in the Bcc field? (1)
   b. What is the domain name of the group to whom the email is being sent? (1)
   c. The email address *designteam@weborg.co.za* is being sent to a mailing list. Explain what is meant by a mailing list. (2)
   d. Will the members of the design team know that the email has been sent to their manager? Explain your answer. (3)
4. Innocent has two email accounts. The email addresses are *innocent25@gmail.com* and *inno.mhlebi@mweb.co.za*.
   a. Which email address is a web-based email account? (1)
   b. List one advantage and one disadvantage of having a web-based email account. (2)
   c. What is the domain name of Innocent's ISP? (1)
5. Amahle is a long-distance runner who participates often in marathons.
   a. Amahle would like to keep an online journal of her experiences. What e-communication application should she use? (1)
   b. Mention two things that she could write about each day. (2)

6. Study the screenshot of an email below and use it to the answer the questions that follow:



   This email does not follow good netiquette.
   a. What is meant by "netiquette"? (2)
   b. Mention three ways in which the writer has not followed good netiquette. (3)

TOTAL: [26]

AT THE END OF THE CHAPTER

Use the checklist to make sure that you worked through the following and that you understand it.

	DID YOU ...	YES	NO
1.	Learn what electronic communication is.		
2.	Identify the different electronic communication devices.		
3.	Learn what ISP and web-based mail are.		
4.	Learn about the advantages and disadvantages of ISP versus webmail.		
5.	Understand the different features of email, such as the “Cc” and “Bcc” fields, attachments, and address books.		
6.	Learn how to compose email messages.		
7.	Understand how to send, receive, forward, reply to and reply to all using email.		
8.	Learn about netiquette over email.		

TERM 3

SOCIAL IMPLICATIONS: EMAIL AND INTERNET

CHAPTER 13

CHAPTER OVERVIEW

Unit 13.1 Social implications: Email and internet safety

At the end of this chapter, you should be able to:

- Identify email threats, issues and prevention methods.
- Use email and the internet safely.

INTRODUCTION

The email and internet are not always safe and secure. You might receive emails from scammers and cybercriminals who are looking to get information from you, such as your banking information and passwords. You should, therefore, avoid storing any sensitive information, such as credit card numbers and passwords on your computer, or sending details to anyone asking for them. In this section, we will look at email threats and what you can do to ensure safe email and internet use.